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# Product Manifold Flow Matching for Mixed-Type Tabular Data Generation

**Shikun Chen**[1] · **Yangguang Liu**[1]

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**Abstract**
Synthetic tabular data generation is increasingly important for privacy-preserving machine learning in healthcare, finance, and other regulated industries. Many current methods embed categorical variables into continuous spaces, which can blur their discrete structure and introduce interpolations between categorical states. Here, we propose flow matching on product manifolds, where each variable type resides on its natural geometric space: continuous variables in Euclidean space, categorical variables on probability simplexes with Fisher-Rao geometry, and ordinal variables on order-preserving arcs of unit circles. Our framework employs coordinate-wise vector field decomposition for manifold-specific processing and geometry-aware midpoint aggregation to capture inter-type dependencies with preserved geometric consistency. We evaluate the method on 12 benchmark datasets that range from small clinical records to educational datasets with over 101 million samples. Results show measurable improvements in generation quality, particularly on categorical-heavy and large-scale datasets. Geometric constraints also exhibit empirical resistance to memorization without explicit noise injection, as measured by distance to the closest record and membership inference metrics. The coordinate-wise structure enables efficient parallel computation that scales well to large datasets.

**Keywords** Flow matching · Product manifolds · Mixed-type data · Tabular data generation · Geometric deep learning

## 1 Introduction

Synthetic tabular data generation has experienced increasing importance across scientific, industrial, and commercial domains. Beyond the inherent capabilities of enabling privacy-preserving data sharing, synthetic data generation provides a complete toolset for address-

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✉ Yangguang Liu
liuyangguang@nbufe.edu.cn

[1] College of Artificial Intelligence, Ningbo University of Finance & Economics, Ningbo 315175, China

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ing data scarcity, regulatory compliance, and collaborative machine learning in sensitive domains such as healthcare and finance (Borisov et al., 2022; Cremaschi et al., 2024; Xu et al., 2019; Liu et al., 2025). The ability to generate high-quality synthetic datasets allows organizations to share valuable insights while protecting sensitive information, enabling actionable decisions in data-constrained environments.

Current approaches face a geometric challenge when modeling mixed-type tabular data. Mixed-type datasets contain both continuous variables that naturally reside in Euclidean space and categorical variables that possess discrete, unordered structure (Shi et al., 2025). Existing generative modeling approaches typically map variables into a single continuous space through techniques such as one-hot encoding or learned embeddings (Xu et al., 2019; Zhao et al., xxx; Kotelnikov et al., 2023). This unified representation enables the application of well-established continuous generative models, but it can introduce geometric distortions that become more visible as dataset scale and complexity increase.

This geometric mismatch manifests in measurable ways. Discrete categorical variables embedded in continuous spaces may pass through smooth interpolations between categorical states that lack meaningful interpretation. During generation, such interpolations can produce intermediate states corresponding to invalid categorical combinations, reducing the preservation of categorical relationships. The continuity imposed on discrete variables can also distort learned probability distributions and affect statistical fidelity (Rashki, 2024; Yan et al., 2025).

Recent advances in generative modeling have witnessed substantial progress in flow matching (Lipman et al., 2022; Albergo & Vanden-Eijnden, 2022; Liu et al., 2022) and diffusion models (Ho et al., 2020) for tabular data synthesis. Methods such as TabDDPM (Kotelnikov et al., 2023), TABSYN (Zhang et al., 2023), and the recent TabbyFlow with exponential family variational flow matching (Guzmán-Cordero et al., 2025) have achieved state-of-the-art performance by developing specialized techniques to handle mixed-type data. These approaches represent notable engineering achievements, employing sophisticated noise schedules, moment matching objectives, and hybrid continuous-discrete diffusion processes. They also highlight a remaining modeling gap: mixed-type tables contain variables with different natural geometries, while many existing objectives still rely on continuous relaxations for all components.

Parallel developments in geometric deep learning have established that respecting the natural geometric structure of data leads to more principled and effective learning algorithms (Gerken et al., 2023). The recognition that different types of data naturally reside on different manifolds – images on pixel grids, graphs on irregular structures, and molecular conformations on Riemannian manifolds – has driven methodological advances, and enforcing geometric consistency has proved similarly beneficial in neighboring domains, from shape correspondence and globally optimal 3D shape matching to generalizable scene reconstruction (Sahillioğlu, 2020; Roetzer & Bernard, 2025; Hosseinzadeh et al., 2025). Recent work in product manifold machine learning has demonstrated the benefits of simultaneously processing data representations across multiple geometric spaces (Woodward et al., 2024), while advances in Riemannian flow matching have shown how geometric principles can enhance generative modeling on non-Euclidean spaces (Miller et al., 2024; Cheng et al., 2024; Lipman et al., 2024).

Here, we explore how geometric flow matching can be incorporated into mixed-type tabular data generation by recognizing that different variable types naturally reside on distinct

geometric spaces. Our approach operates directly on product manifolds where continuous variables remain in their natural Euclidean space while categorical variables reside on statistical manifolds equipped with appropriate geometric structure. We evaluate the resulting model across diverse datasets from small clinical records to massive educational datasets with over 101 million samples.

We develop geometric flow matching on product manifolds for mixed-type tabular data generation, which we refer to as Product-Manifold Flow Matching (PMFM). The framework builds on established tools from Riemannian geometry and flow matching and makes the following contributions:

- We combine product manifold flow matching with Fisher-Rao geometry for categorical variables, coordinate-wise decomposition, and a circular treatment that covers periodic and monotone ordinal variables alike, into a unified generative framework for mixed-type tabular data. Individual geometric components such as Fisher-Rao geodesics (Davis et al., 2024; Cheng et al., 2024) and product manifold decomposition (Chen & Lipman, 2023) have been studied separately; PMFM integrates them for heterogeneous tabular generation.
- We introduce geometry-aware midpoint aggregation, a geometrically consistent mechanism for modeling inter-type dependencies across manifold boundaries (Definition 4), which combines fixed-base tangent-space averaging with a mutual-information-based dependency graph. Ablations compare this mechanism with simple concatenation and standard cross-attention conditioning (Sect. 4.7).
- We evaluate across 12 benchmark datasets spanning from 303 to 101 million samples, showing that geometric consistency provides measurable benefits on categorical-heavy and large-scale datasets, while being competitive elsewhere. We provide computational efficiency comparisons, correlation preservation analysis, and privacy evaluation using geometrically consistent metrics.

The coordinate-wise vector field decomposition and loss decomposition on product manifolds follow from standard properties of product Riemannian manifolds (7.3, 7.4). Our contribution is in applying these properties to construct an efficient and geometrically consistent framework for mixed-type tabular data. Our proposed framework is shown in Fig. 1.

### 1.1 Related Work

Synthetic tabular data generation has progressed through multiple methodological stages, each offering a different perspective on how to model datasets that combine continuous and categorical variables. Early approaches based on traditional machine learning relied on statistical formulations and tree-based techniques. These methods often required explicit distributional assumptions or discretization procedures, which simplified the learning problem but could distort the true data structure (Sklar, 1973; Chawla et al., 2002; Han et al., 2005; Wang et al., 2023; Mukherjee & Khushi, 2021; He et al., 2008; Nowok et al., 2016). Variational autoencoder (VAE) frameworks later introduced latent variable modeling to tabular domains, enabling compressed representations for sample generation, although issues such as posterior collapse and the sparsity induced by categorical encodings frequently limited performance (Liu et al., 2023; Ma et al., 2020; Xu et al., 2019).

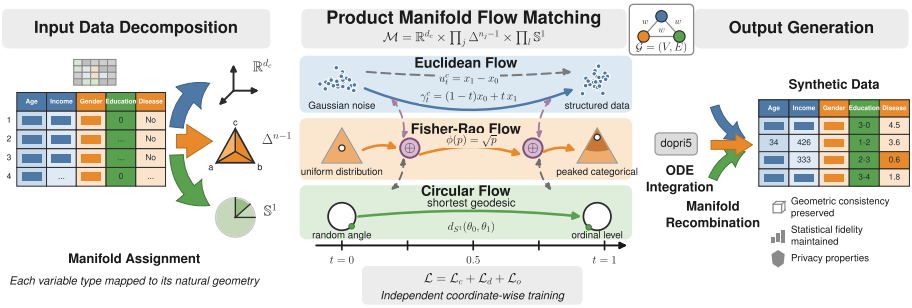



**Fig. 1** Overview of our proposed geometric flow matching framework for mixed-type tabular data. The framework operates on product manifolds $\mathcal{M} = \mathbb{R}^{d_c} \times \prod_{j=1}^{k} \Delta^{n_j-1} \times \prod_{l=1}^{m} S^1$, where continuous variables reside in Euclidean space, categorical variables on probability simplexes with Fisher-Rao geometry, and ordinal variables on order-preserving arcs of unit circles. The approach enables coordinate-wise flow matching on each manifold component while handling inter-type dependencies through geometry-aware midpoint aggregation operations. This preserves the intrinsic geometric structure of each data type while enabling unified generative modeling of heterogeneous tabular data

Generative adversarial networks (GANs) marked another major development by introducing adversarial training and specialized architectures with mode-aware normalization and conditional generation mechanisms. These methods often represent discrete variables through continuous encodings, which motivates geometric treatments of categorical structure (Xu & Veeramachaneni, 2018; Choi et al., 2017; Alshantti et al., 2024; Zhao et al., 2022, xxx; Engelmann & Lessmann, 2021; Zhao et al., 2024; Torfi et al., 2022). Diffusion-based models offered a more principled alternative by treating continuous and categorical variables through separate stochastic processes, typically Gaussian diffusion for the former and multinomial diffusion for the latter. This strategy implicitly acknowledged the product-space structure of tabular data, while leaving open how to represent continuous, categorical, and ordinal components within a single geometric framework (Kotelnikov et al., 2023; Zhang et al., 2023; Shi et al., 2024; Lin et al., 2024; Suh et al., 2023; He et al., 2024; Sattarov et al., 2023; Liu & Liu, 2024; Jolicoeur-Martineau et al., 2024).

Large language model (LLM)-based approaches have more recently been positioned as an alternative paradigm (Kim et al., 2024; Seedat et al., 2023; Yang et al., 2024; Borisov et al., 2022; Zhao et al., 2025). By treating tabular records as token sequences, they leverage pre-trained language representations for generative synthesis. Purely numerical relationships remain difficult to model within this framework, and the computational overhead of training and inference can be prohibitive in practical deployments.

Flow matching has recently gained prominence as a powerful paradigm for generative modeling, achieving state-of-the-art results across a wide range of domains, from images and videos to speech, proteins, and robotics (Lipman et al., 2022; Esser et al., 2024; Le et al., 2023; Vyas et al., 2023; Huguet et al., 2024; Black et al., 2024). At its core, the method learns velocity fields that define trajectories through ordinary differential equations (ODEs), thereby transporting samples from a source distribution to a target distribution via deterministic and continuous bijections. This formulation is highly flexible and has been extended beyond classical Euclidean spaces. Examples include Riemannian flow matching on general

geometries (Chen & Lipman, 2023), variants for manifold-valued protein structures (Yim et al., 2023), and generator matching for arbitrary Markov processes (Holderrieth et al., 2024).

Motivated by these developments, several works have explored flow-based approaches for tabular data generation. TabbyFlow (Guzmán-Cordero et al., 2025), for instance, introduces exponential-family variational flow matching to align mixed-type variables through moment matching, whereas Forest Flow (Jolicoeur-Martineau et al., 2024) demonstrates that gradient-boosted trees can serve as effective estimators of velocity fields. A parallel line of work places categorical variables on statistical manifolds endowed with Fisher–Rao geometry (Cheng et al., 2024): Fisher Flow Matching (Davis et al., 2024) maps the simplex $\Delta^{n-1}$ to the positive orthant of the unit hypersphere $\mathbb{S}_+^{n-1}$ via the diffeomorphism $\phi(p) = \sqrt{p}$, enabling geodesic interpolation on the statistical manifold, and is effective for purely categorical data in applications such as DNA sequence generation, molecular design, and language modeling. From this line of work PMFM inherits the Fisher–Rao simplex geometry with its spherical parameterization (Cheng et al., 2024; Davis et al., 2024) and the geodesic construction of Riemannian conditional flow matching (Lipman et al., 2024; Chen & Lipman, 2023), and adapts them by running geodesic flow matching per factor on a mixed product manifold rather than on a single homogeneous space. New to PMFM are the mixed-type product construction itself, whose per-variable manifold assignment includes order-preserving arcs for ordinal variables, together with the geometry-aware midpoint aggregation (Definition 4) that carries dependencies across manifold boundaries.

Table 1 situates PMFM among the closest geometric and tabular flow-matching methods. Two elements distinguish the construction: each variable type is assigned its native manifold, including an order-preserving circular geometry for ordinal variables, and cross-type dependencies are modeled by a geometrically consistent operation on the resulting product manifold. Geometric flow matching such as Fisher Flow operates on a single manifold for discrete data, whereas mixed-type tabular generators such as TabbyFlow and Forest Flow handle heterogeneity through continuous relaxation or tree heuristics.

**Table 1** Positioning of PMFM relative to representative geometric and tabular flow-matching methods

| Method | Continuous | Categorical | Ordinal | Cross-type dependency | Target data |
|---|---|---|---|---|---|
| Riemannian FM (Lipman et al., 2024; Chen & Lipman, 2023) | manifold geodesics | – | – | – | single-manifold |
| Fisher Flow (Davis et al., 2024) | – | Fisher–Rao simplex | – | – | discrete |
| Categorical FM (Cheng et al., 2024) | – | statistical manifold | – | – | categorical |
| TabbyFlow (Guzmán-Cordero et al., 2025) | exp. family | exp. family | continuous relaxation | moment matching | mixed tabular |
| Forest Flow (Jolicoeur-Martineau et al., 2024) | tree field | tree field | tree field | implicit (tree splits) | mixed tabular |
| PMFM (ours) | Euclidean | Fisher–Rao simplex | circular arc $S^1$ | geometry-aware midpoint | mixed tabular |

PMFM equips each variable type with its native manifold and models cross-type dependencies through a geometrically consistent operation on the product manifold. The ordinal column records the space each method uses; PMFM's arc covers periodic and monotone variables with one construction (Sect. 4.3)

## 2 Preliminaries

We briefly review the components on which our method builds. For additional background on Riemannian flow matching, see 6. Table 2 summarizes the notation used throughout.

### 2.1 Flow Matching

Flow matching (Lipman et al., 2022; Albergo & Vanden-Eijnden, 2022; Liu et al., 2022) learns a time-dependent vector field $u_t : [0,1] \times \mathcal{X} \to T\mathcal{X}$ that transports a source distribution $p_0$ to a target distribution $p_1$ while satisfying the continuity equation $\frac{\partial p_t}{\partial t} + \nabla \cdot (p_t u_t) = 0$. Conditional flow matching (CFM) avoids direct probability path modeling by constructing conditional paths $p_t(x|x_1)$ for data pairs $x_0 \sim p_0, x_1 \sim p_1$. Using interpolants $x_t = tx_1 + (1-t)x_0$

**Table 2** Notation used throughout the paper

| Symbol | Meaning |
|---|---|
| $\mathcal{M}$ | Product manifold $\mathbb{R}^{d_c} \times \prod_j \Delta^{n_j-1} \times \prod_l S^1$ |
| $d_c$, $k$, $m$ | Number of continuous, categorical, ordinal variables |
| $R$ | Number of component manifolds of $\mathcal{M}$; factor index $r$ |
| $N$ | Number of samples in the dataset |
| $n_j$ | Number of categories of the $j$-th categorical variable |
| $\Delta^{n-1}$ | Probability simplex over $n$ categories |
| $S^1$ | Unit circle; monotone ordinals use order-preserving arcs |
| $d_{S^1}$ | Geodesic distance on the unit circle |
| $\phi$ | Diffeomorphism $\phi(p) = \sqrt{p} : \Delta^{n-1} \to \mathbb{S}^{n-1}_+$ |
| $d_{FR}$ | Fisher–Rao distance on the simplex |
| $\|v\|_{FR,p}$ | Fisher–Rao norm of tangent vector $v$ at base point $p$ |
| $x_t^c$, $x_t^{d,j}$, $x_t^{o,l}$ | Component states at flow time $t$ (subscript: time; superscript: variable) |
| $x^{\tau,i}$, $\tilde{x}^{\tau,i}$ | Generic variable of type $\tau$ and its dependency-aggregated representation |
| $u_t$, $u_\theta$ | Target and learned time-dependent vector field |
| $u_t^{circ}$ | Conditional circular velocity field (Eq. 10) |
| $\gamma_t$ | Geodesic interpolation path |
| $\exp_\mu$, $\log_\mu$ | Riemannian exponential and logarithmic maps at $\mu$ |
| $\mu$ | Reference point on the simplex (uniform distribution) |
| $\tau$, $\tau'$ | Variable-type indices, $\tau \in \{c, d, o\}$ |
| $\sigma$ | Logistic sigmoid function |
| $\Pi_{\tau \leftarrow \tau'}$ | Geometric projection from a type-$\tau'$ variable to a type-$\tau$ variable |
| $G = (V, E)$ | Inter-type dependency graph, vertices, edges |
| $w_{ij}$, $W$ | Dependency edge weight; weighted adjacency matrix |
| $\rho$ | Edge-weight threshold ($\rho = 0.1$) |
| $\alpha$ | Aggregation strength |
| $[s]$ | Aggregation iteration index (distinct from flow time $t$) |
| $S$ | Number of aggregation passes that build the context ($S = 2$) |
| $a_{ij}$, $b_{ij}$, $c_{ij}$ | Learned parameters of the projection maps of edge $(i \leftarrow j)$ (Eq. 16) |
| $L_\Pi$ | Lipschitz bound of the projection operators |
| $C_\Theta$ | Chart-distortion constant of the fixed-base simplex aggregation (Theorem 1) |
| $\eta$ | Learning rate |
| $p_0$, $p_1$ | Source and target distributions |

, the marginal vector field is recovered as $u_t(x) = \mathbb{E}_{x_1 \sim p_1}[u_t(x|x_1) \mid x_t = x]$, and training reduces to regression:

$$\mathcal{L}_{CFM}(\theta) = \mathbb{E}_{t,x_1,x_t} \|u_\theta(t, x_t) - u_t(x_t|x_1)\|^2 \tag{1}$$

### 2.2 Product Manifolds

Given Riemannian manifolds $(\mathcal{M}_1, g_1), \ldots, (\mathcal{M}_R, g_R)$, the product manifold $\mathcal{M} = \mathcal{M}_1 \times \cdots \times \mathcal{M}_R$ has tangent space decomposition $T_x\mathcal{M} = T_{x^1}\mathcal{M}_1 \oplus \cdots \oplus T_{x^R}\mathcal{M}_R$ and product metric $g_x(u, v) = \sum_{r=1}^{R} g_{x^r}(u^r, v^r)$. This structure enables coordinate-wise geometric operations, so that exponential maps, logarithmic maps, and geodesics can be computed independently on each component.

For mixed-type tabular data, the natural product structure is $\mathcal{M} = \mathbb{R}^{d_c} \times \prod_{j=1}^{k} \Delta^{n_j-1} \times \prod_{l=1}^{m} S^1$, where continuous variables reside in Euclidean space $\mathbb{R}^{d_c}$, categorical variables with $n_j$ classes on probability simplexes $\Delta^{n_j-1} = \{p \in \mathbb{R}^{n_j} : p_i \geq 0, \sum_i p_i = 1\}$, and ordinal variables on circular arcs of the unit circle $S^1$ under an arc embedding that preserves their ordering (Sect. 3).

Throughout the paper we say that an operation *preserves manifold structure* when it satisfies three requirements: (i) every state it produces lies on $\mathcal{M}$, so no intermediate sample leaves the valid data space; (ii) interpolation between states follows geodesics of the intrinsic metric rather than straight lines of an ambient embedding; and (iii) discrepancies between vector fields are measured with the Riemannian metric of each factor. Linear interpolation of one-hot encodings illustrates what is lost otherwise: the straight segment between two simplex vertices stays inside the simplex but is not a Fisher–Rao geodesic, so equal parameter increments do not correspond to equal statistical distances, and the induced paths misrepresent how categorical distributions deform into one another.

Each factor carries an explicit metric, which we fix here once for the whole paper. The Euclidean factor uses the norm $\|\cdot\|_2$. The simplex factor uses the Fisher–Rao geometry of Sect. 2.3. The circle $S^1$ carries the geodesic distance $d_{S^1}(\theta_1, \theta_2) = \min(|\theta_1 - \theta_2|,\ 2\pi - |\theta_1 - \theta_2|)$, and its tangent spaces $T_\theta S^1 \cong \mathbb{R}$ carry the absolute-value norm $|v|$.

### 2.3 Fisher-Rao Geometry on Probability Simplexes

Probability distributions possess intrinsic geometric structure through the Fisher information metric. For distributions $p, q \in \Delta^{n-1}$, the Fisher-Rao distance

$$d_{FR}(p, q) = 2 \arccos\left(\sum_{i=1}^{n} \sqrt{p_i q_i}\right) \tag{2}$$

quantifies statistical distinguishability through the Fisher-Rao metric. The corresponding Riemannian metric on the tangent space $T_p\Delta^{n-1} = \{v \in \mathbb{R}^n : \sum_i v_i = 0\}$ is the Fisher information metric, whose squared norm we write as $\|v\|^2_{FR,p} = \sum_{i=1}^{n} v_i^2/p_i$; the base point $p$ appears explicitly in this notation, and $\|\cdot\|_2$ remains reserved for the Euclidean norm. The diffeomorphism $\phi : \Delta^{n-1} \to \mathbb{S}^{n-1}_+$ defined by $\phi(p) = \sqrt{p}$ maps the simplex to the posi-

tive orthant of the unit sphere (since $|\phi(p)|^2 = \sum_i p_i = 1$), enabling geodesic computation through spherical geometry.

**Proposition 1** *(Fisher-Rao Geodesic) The Fisher-Rao geodesic connecting $p, q \in \Delta^{n-1}$ is:*

$$\gamma(t) = \phi^{-1}\left(\frac{\sin((1-t)\theta)\sqrt{p} + \sin(t\theta)\sqrt{q}}{\sin(\theta)}\right) \tag{3}$$

*where $\theta = \arccos\left(\sum_{i=1}^{n} \sqrt{p_i q_i}\right)$ is the spherical angle between $\sqrt{p}$ and $\sqrt{q}$ (note $d_{FR}(p,q) = 2\theta$), and $\phi^{-1}(v) = v \odot v$. This geodesic satisfies $\gamma(0) = p$, $\gamma(1) = q$, and $\gamma(t) \in \Delta^{n-1}$ for all $t \in [0,1]$.*

***Proof*** See 7.1.

Proposition 1 supplies the closed-form interpolant that our training procedure evaluates at arbitrary times $t$, and its constant-speed property means that equal increments of $t$ traverse equal Fisher–Rao distances. Unlike Euclidean linear interpolation $(1-t)p + tq$, which can produce geometrically unnatural paths on the simplex, Fisher-Rao geodesics respect the manifold structure inherently, preserving statistical coherence throughout the interpolation. Riemannian flow matching (Lipman et al., 2024; Chen & Lipman, 2023) extends CFM to general manifolds by replacing linear interpolation with geodesic interpolation and standard vector operations with their Riemannian counterparts. On product manifolds, this decomposition operates coordinate-wise: each component follows geodesics on its own manifold while preserving joint probability structure (see 6).

## 3 Geometric Flow Matching on Product Manifolds

### 3.1 Motivation and Framework

PMFM operates directly on the natural product manifold structure of mixed-type tabular data, avoiding a single continuous embedding for heterogeneous variable types.

**Definition 1** (Mixed-Type Tabular Data Representation) A mixed-type tabular dataset consists of $N$ samples, each of which decomposes as:

$$x = (x^c, x^d, x^o) \in \mathbb{R}^{d_c} \times \prod_{j=1}^{k} \Delta^{n_j - 1} \times \prod_{l=1}^{m} S^1$$

where $x^c \in \mathbb{R}^{d_c}$ represents continuous features, $x^d = (x^{d,1}, \ldots, x^{d,k})$ represents categorical features with $x^{d,j} \in \Delta^{n_j-1}$, and $x^o = (x^{o,1}, \ldots, x^{o,m})$ represents ordinal features with $x^{o,l} \in S^1$ placed through the arc embedding of Eq. 4, which preserves ordering relationships. A raw category label $c \in \{1, \ldots, n_j\}$ enters the model as the smoothed one-hot vector $x^{d,j} = (1-\varepsilon)\, e_c + (\varepsilon/n_j)\, 1$ with $\varepsilon = 10^{-2}$, which places every observation in the interior of the simplex where the geodesic formulas below are smooth.

An ordinal variable with levels $\{1, \dots, K\}$ is placed on $S^1$ through the arc embedding

$$\theta_k = \pi \frac{k-1}{K-1}, \qquad k = 1, \dots, K, \tag{4}$$

which spans the half-circle $[0, \pi]$. The geodesic distance on $S^1$ between levels $a$ and $b$ is then $d_{S^1}(\theta_a, \theta_b) = \pi\,|a-b|/(K-1)$, increasing monotonically with the level gap and never wrapping, so $d(1,2) < d(1,4) < d(1,5)$ holds on every scale. The half-arc is the unit interval embedded isometrically in $S^1$ up to the fixed scale $\pi$, which lets periodic and monotone ordinal variables share one manifold type, one pair of projection operators, and one aggregation rule. Genuinely periodic ordinals, such as hour of day or month, instead use the full circle with $\theta_k = 2\pi(k-1)/K$, recovering the cyclic adjacency appropriate to those variables. In both cases the circular geometry supplies the manifold structure used by the flow-matching and aggregation operations while matching the ordering semantics of the data. Section 4.3 compares the half-arc, interval, and full-circle encodings under matched model capacity, on both real and controlled synthetic ordinal data.

**Definition 2** (Product Manifold Flow Matching) For the product manifold $\mathcal{M} = \mathbb{R}^{d_c} \times \prod_{j=1}^{k} \Delta^{n_j-1} \times \prod_{l=1}^{m} S^1$, we learn a vector field whose tangent-space decomposition is

$$u_t(x) = (u_t^c(x), u_t^{d,1}(x), \dots, u_t^{d,k}(x), u_t^{o,1}(x), \dots, u_t^{o,m}(x))$$

where $u_t^c$ takes values in $\mathbb{R}^{d_c}$ and employs Euclidean flow matching, $u_t^{d,j}$ takes values in $T\Delta^{n_j-1}$ and employs Fisher-Rao flow matching, and $u_t^{o,l}$ takes values in $TS^1$ and employs circular flow matching preserving ordinal structure. The decomposition constrains where each component's values live; each component may depend on the full state $x$ through the shared aggregated context of Sect. 3.4.

The coordinate-wise decomposition enables independent geometric operations on each manifold component: Euclidean linear interpolation for continuous variables, Fisher-Rao geodesic interpolation for categorical variables, and circular geodesic interpolation for ordinal variables. This preserves the intrinsic geometric structure of each data type while maintaining computational efficiency through manifold-specific optimization.

Inter-type dependencies are captured through geometry-aware midpoint aggregation operations (Definition 4) that respect manifold boundaries without violating geometric constraints. The framework maintains probabilistic coherence across heterogeneous variable types while enabling tractable flow matching through coordinate-wise decomposition. Algorithm 1 formalizes our complete approach.

**Input:** Dataset $\mathcal{D}$ ($N$ samples), dependency graph $G = (V, E)$, weights $w_{ij}$, strength $\alpha$, passes $S$
**Output:** Trained model $u_\theta$ for synthetic data generation
**for** epoch = 1 to $N_{epochs}$ **do**
  **for** batch $\mathcal{B}$ in dataset **do**
    Sample $t \sim \mathcal{U}[0,1]$, $x_1 \sim \mathcal{B}$, $x_0 \sim p_0 = \mathcal{N}(0, I) \times \prod_j \mathrm{Dir}(\mathbf{1}) \times \prod_l \mathcal{U}[0, \pi]$
    // **Coordinate-wise geodesic interpolation**
    $x_t^c \leftarrow (1-t)x_0^c + tx_1^c$ {Euclidean interpolation}
    **for** $j = 1$ to $k$ **do**
      $x_t^{d,j} \leftarrow \mathrm{FisherRaoGeodesic}(x_0^{d,j}, x_1^{d,j}, t)$ {Eq. 3}
    **end for**
    **for** $l = 1$ to $m$ **do**
      $x_t^{o,l} \leftarrow \mathrm{CircularGeodesic}(x_0^{o,l}, x_1^{o,l}, t)$ {Eq. 9}
    **end for**
    // **Compute target vector fields**
    $u_t^{c,target} \leftarrow x_1^c - x_0^c$ {Euclidean velocity}
    **for** $j = 1$ to $k$ **do**
      $u_t^{d,j,target} \leftarrow \mathrm{FisherRaoVelocity}(x_0^{d,j}, x_1^{d,j}, t)$ {Eq. 5}
    **end for**
    **for** $l = 1$ to $m$ **do**
      $u_t^{o,l,target} \leftarrow \mathrm{CircularVelocity}(x_0^{o,l}, x_1^{o,l}, t)$ {Eq. 10}
    **end for**
    // **Dependency aggregation on the graph:** $S$ **passes of Eq. 22**
    $\tilde{x}_t^{[0]} \leftarrow x_t$
    **for** $s = 1$ to $S$ **do**
      **for** each variable $(\tau, i) \in V$ **do**
        $\tilde{x}_t^{\tau,i,[s]} \leftarrow \mathrm{GeoMidpoint}\big(\tilde{x}_t^{\tau,i,[s-1]}, \{\Pi_{\tau \leftarrow \tau'}(\tilde{x}_t^{\tau',j,[s-1]})\}_{(\tau',j)\in N(i)}; 1-\alpha, \{\alpha w_{ij}\}\big)$
      **end for**
    **end for**
    $\tilde{x}_t \leftarrow \tilde{x}_t^{[S]}$ {$S = 2$ in all experiments}
    // **Predict vector fields from state and aggregated context**
    $(u_t^{c,pred}, \{u_t^{d,j,pred}\}_{j=1}^k, \{u_t^{o,l,pred}\}_{l=1}^m) \leftarrow u_\theta(t, x_t, \tilde{x}_t)$
    // **Compute coordinate-wise loss**
    $\mathcal{L}_c \leftarrow \|u_t^{c,pred} - u_t^{c,target}\|_2^2$
    $\mathcal{L}_d \leftarrow \sum_{j=1}^k \|u_t^{d,j,pred} - u_t^{d,j,target}\|^2_{FR,x_t^{d,j}}$
    $\mathcal{L}_o \leftarrow \sum_{l=1}^m |u_t^{o,l,pred} - u_t^{o,l,target}|^2$
    // **Update parameters**
    $\theta \leftarrow \theta - \eta \nabla_\theta(\mathcal{L}_c + \mathcal{L}_d + \mathcal{L}_o)$ {Learning rate $\eta$}
  **end for**
**end for**
// **Generation process**
**for** sample = 1 to $N_{samples}$ **do**
  Sample $x_0 \sim p_0$ {Same source distribution as training}
  $x_1 \leftarrow \mathrm{ODESolve}(u_\theta, x_0, [0,1])$ {$S$ aggregation passes inside each evaluation}
  Decode: $\hat{x}^{d,j} \leftarrow \arg\max_c x_1^{d,j}$, $\hat{x}^{o,l} \leftarrow$ nearest ordinal level to $x_1^{o,l}$
**end for**

**Algorithm 1** Geometric flow matching on product manifolds

This algorithm captures our complete geometric flow matching framework, incorporating coordinate-wise flow matching across continuous (Euclidean), categorical (Fisher-Rao), and ordinal (circular) manifolds, with geometry-aware midpoint aggregation (Definition 4) preserving geometric consistency across heterogeneous variable types. Training and generation use the same transport: both draw $x_0$ from the per-factor maximum-entropy priors of Sect. 3.5, so the field learned from prior-to-data pairs is exactly the field integrated at inference, and the dependency aggregation of Sect. 3.4 enters both phases through the context $\tilde{x}_t$.

The algorithmic functions are defined as follows: `FisherRaoGeodesic` implements Proposition 1, `FisherRaoVelocity` computes Eq. 5, `CircularGeodesic` and `CircularVelocity` implement Eqs. 9 and 10, and `GeoMidpoint` implements Definition 4 with the coordinate-wise operations from Eq. 12, applied for $S$ passes as in Eq. 22.

## 3.2 Fisher-Rao Geodesic Flow Matching

Categorical variables require specialized geometric treatment to preserve their discrete probabilistic structure during flow matching. Using the Fisher-Rao geodesic from Proposition 1, we construct a conditional flow matching velocity field for categorical variables on probability simplexes.

For conditional flow matching between categorical distributions $p_0$ and $p_1$, we derive the target velocity field from the geodesic formula (Eq. 3). Writing $\xi = \sqrt{p_0}$, $\zeta = \sqrt{p_1}$, and noting that the Fisher-Rao geodesic satisfies $\gamma(t) = [\gamma_{sphere}(t)]^{\odot 2}$ (element-wise squaring of the spherical geodesic), we apply the chain rule to obtain:

$$u_t^{FR}(p|p_1) = \frac{d}{dt}\gamma(t)\Big|_{t:\gamma(t)=p} = \begin{cases} \frac{2\theta}{\sin(\theta)} \cdot \sqrt{p} \odot [-\cos((1-t)\theta)\xi + \cos(t\theta)\zeta] & \text{if } \theta > \epsilon \\ 2\sqrt{p} \odot (\zeta - \xi) & \text{if } \theta \leq \epsilon \end{cases} \tag{5}$$

where $\odot$ denotes element-wise (Hadamard) multiplication, $\epsilon = 10^{-8}$ prevents singularities when distributions are nearly identical, and the second case uses the limit as $\theta \to 0$ where $\lim_{\theta \to 0} \frac{\theta}{\sin(\theta)} = 1$. A detailed step-by-step derivation of this velocity field formula from the geodesic equation is provided in 7.2.

This formulation constrains Fisher–Rao geodesics to the probability simplex throughout the interpolation process, thereby maintaining the probabilistic interpretation of categorical variables. The spherical parameterization eliminates singularities near simplex boundaries and ensures stable gradient computation during training.

The Fisher-Rao approach (Proposition 1) integrates seamlessly into our coordinate-wise framework: categorical variables follow Fisher-Rao geodesics while continuous variables follow Euclidean paths, with both contributing to the joint probability transport on the product manifold without geometric inconsistencies.

### 3.3 Coordinate-Wise Vector Field Construction

The tangent space of a product manifold decomposes as $T_x\mathcal{M} = T_{x^1}\mathcal{M}_1 \oplus \cdots \oplus T_{x^R}\mathcal{M}_R$, so any vector field $u_t : \mathcal{M} \to T\mathcal{M}$ decomposes uniquely into components $u_t(x) = (u_t^1(x), \ldots, u_t^R(x))$ with $u_t^r(x) \in T_{x^r}\mathcal{M}_r$, which is a standard property of product Riemannian manifolds (see 7.3 for a formal statement). The decomposition constrains where the values of each component live, not which arguments it takes. Restricting the $r$-th component to the $r$-th coordinate is an architectural choice; PMFM instead passes every component field the shared aggregated context of Sect. 3.4, so cross-column dependence, which is central to tabular data, enters each one, with the coordinate-restricted alternative evaluated as Ours-NoAgg (Sect. 4.7). Factor curvature does not affect this: the Fisher–Rao simplex is isometric to a positively curved sphere orthant and $S^1$ is flat but topologically non-trivial (Luo et al., 2024), while the product metric and tangent structure still split factor-wise (Lipman et al., 2024; Chen & Lipman, 2023). For our mixed-type tabular data on $\mathcal{M} = \mathbb{R}^{d_c} \times \prod_{j=1}^{k} \Delta^{n_j-1} \times \prod_{l=1}^{m} S^1$, this yields:

$$u_t(x) = (u_t^c(x), \{u_t^{d,j}(x)\}_{j=1}^k, \{u_t^{o,l}(x)\}_{l=1}^m) \tag{6}$$

Similarly, the product metric $g_x(u, v) = \sum_r g_{x^r}(u^r, v^r)$ implies that the conditional flow matching objective decomposes coordinate-wise ( 7.4):

$$\mathcal{L}_{CFM}(\theta) = \sum_{r=1}^{R} \mathbb{E}_{t,x_1,x_t}\left[\|u_\theta^r(t, x_t) - u_t^r(x_t^r|x_1^r)\|_{g_{x^r}}^2\right] \tag{7}$$

For our mixed-type tabular data, this yields three groups of objectives:

$$\begin{aligned}
\mathcal{L}_c(\theta) &= \mathbb{E}_{t,x_1,x_t}\left[\|u_\theta^c(t,x_t) - (x_1^c - x_0^c)\|_2^2\right] \\
\mathcal{L}_d(\theta) &= \sum_{j=1}^{k} \mathbb{E}_{t,x_1,x_t}\left[\|u_\theta^{d,j}(t,x_t) - u_t^{FR}(x_t^{d,j}|x_1^{d,j})\|^2_{FR,x_t^{d,j}}\right] \\
\mathcal{L}_o(\theta) &= \sum_{l=1}^{m} \mathbb{E}_{t,x_1,x_t}\left[|u_\theta^{o,l}(t,x_t) - u_t^{circ}(x_t^{o,l}|x_1^{o,l})|^2\right]
\end{aligned} \tag{8}$$

For ordinal variables the interpolation and target field mirror the Fisher–Rao case. Writing $\mathrm{w}(\delta)$ for the wrap of $\delta$ to $(-\pi, \pi]$, the circular geodesic between $\theta_0, \theta_1 \in S^1$ is

$$\gamma_t^{circ}(\theta_0, \theta_1) = \theta_0 + t\,\mathrm{w}(\theta_1 - \theta_0), \tag{9}$$

and the conditional circular velocity, constant in $t$ along this path, is

$$u_t^{circ}(\theta \mid \theta_1) = \mathrm{w}(\theta_1 - \theta_0) \in T_\theta S^1 \cong \mathbb{R}. \tag{10}$$

For monotone ordinals on the half-arc $[0, \pi]$ the wrap is never active, since $|\theta_1 - \theta_0| \le \pi$, and the geodesic reduces to linear interpolation of angles.

Here $\|\cdot\|_2$ denotes the Euclidean norm, $\|\cdot\|_{FR,p}$ the Fisher–Rao tangent norm at the current base point (Sect. 2.3), and $|\cdot|$ the norm on $T_\theta S^1$. The additive structure of these objectives enables efficient parallel computation: each component flow respects its natural geometry, with Euclidean linear interpolation $\gamma_t^c = (1-t)x_0^c + tx_1^c$ for continuous variables, Fisher-Rao geodesics with $\gamma_t^d \in \Delta^{n-1}$ for categorical variables, and circular geodesics for ordinal variables.

### 3.4 Product Manifold Aggregation for Inter-type Dependencies

Real-world tabular data exhibits complex inter-type dependencies where continuous, categorical, and ordinal variables are not independent but exhibit statistical and semantic relationships. Traditional approaches either ignore these dependencies or handle them through continuous relaxations that violate geometric structure. Our framework addresses inter-type dependencies through geometry-aware midpoint aggregation (Definition 4) that respect manifold boundaries while enabling information transfer across heterogeneous variable types.

The challenge lies in defining mathematically principled operations that combine information from different geometric spaces without violating the intrinsic structure of each manifold component. We achieve this through fixed-base tangent-space aggregation: information is averaged in a tangent space anchored at a fixed reference point of each factor, which yields closed-form, parallelizable updates while keeping every result on its manifold.

**Definition 3** (Inter-type Dependency Graph) For mixed-type tabular data on product manifold $\mathcal{M} = \mathbb{R}^{d_c} \times \prod_{j=1}^{k} \Delta^{n_j-1} \times \prod_{l=1}^{m} S^1$, the inter-type dependency graph $G = (V, E)$ consists of:

- Vertices $V = V_c \cup V_d \cup V_o$ representing continuous, categorical, and ordinal variables
- Edges $E \subseteq V \times V$ representing statistical dependencies between variable types

- Edge weights $w_{ij} \in [0, 1]$ quantifying dependency strength between variables $i$ and $j$

The edge weights $w_{ij}$ are computed from the training data using normalized mutual information for mixed-type variable pairs, which measures statistical dependence while remaining invariant to variable type. For continuous-continuous pairs, we use Pearson correlation; for categorical-categorical pairs, we use Cramér's V statistic; and for mixed pairs, we discretize continuous variables into deciles (ten quantile bins) before computing mutual information. All weights are normalized to [0, 1] and thresholded at $\rho = 0.1$ to construct a sparse dependency graph that retains only statistically significant dependencies. Because these association measures are symmetric, $G$ is an undirected weighted graph whose adjacency matrix $W$ may contain cycles; this matrix enters the convergence guarantee of Theorem 1 directly through its spectral norm $\|W\|_2$.

The dependency graph captures semantic relationships such as continuous age influencing categorical health status, or ordinal education level correlating with continuous income. These relationships require geometric operations that transfer information across manifold boundaries while preserving the mathematical structure of each component space.

**Definition 4** (Geometry-Aware Midpoint Aggregation on Product Manifolds) For points $z_1, \ldots, z_A$ on product manifold $\mathcal{M} = \mathcal{M}_1 \times \cdots \times \mathcal{M}_R$ with weights $w_1, \ldots, w_A$, the geometry-aware midpoint is defined factor-wise as:

$$\begin{aligned}\mathrm{Mid}(z_1, \ldots, z_A; w_1, \ldots, w_A) = \Big(&\mathrm{Mid}_{\mathcal{M}_1}(z_1^1, \ldots, z_A^1; w_1, \ldots, w_A), \ldots, \\ &\mathrm{Mid}_{\mathcal{M}_R}(z_1^R, \ldots, z_A^R; w_1, \ldots, w_A)\Big)\end{aligned} \tag{11}$$

where $\mathrm{Mid}_{\mathcal{M}_r}$ denotes the factor-specific midpoint operation of Eq. 12.
For our mixed-type data, this yields coordinate-wise midpoint operations:

$$\mathrm{Mid}_{\mathbb{R}^{d_c}}(z_1^c, \ldots, z_A^c; w_1, \ldots, w_A) = \sum_{a=1}^{A} w_a z_a^c \tag{12}$$

$$\mathrm{Mid}_{\Delta^{n-1}}(p_1, \ldots, p_A; w_1, \ldots, w_A) = \exp_\mu\left(\sum_{a=1}^{A} w_a \log_\mu(p_a)\right) \tag{13}$$

$$\mathrm{Mid}_{S^1}(\theta_1, \ldots, \theta_A; w_1, \ldots, w_A) = \begin{cases} \sum_{a=1}^{A} w_a \theta_a & \text{(monotone ordinals, arc } [0, \pi]) \\ \arg\left(\sum_{a=1}^{A} w_a e^{i\theta_a}\right) & \text{(periodic ordinals)} \end{cases} \tag{14}$$

where $\mu$ is a fixed reference point on the probability simplex, taken as the uniform distribution. Each factor uses the mean that admits a closed form on its geometry. The Euclidean factor uses the weighted arithmetic mean, which coincides with the intrinsic mean of a flat space for weights summing to one, and for monotone ordinals the arc is isometric to an interval, so the weighted average of angles is again the exact intrinsic mean. The simplex factor averages in the tangent space at the fixed base $\mu$ via $\log_\mu$ and $\exp_\mu$, a fixed-base surrogate for the intrinsic mean requiring one pair of maps per update rather than an inner

optimization loop, and periodic ordinals use the extrinsic resultant-vector mean, whose well-definedness is quantified after Theorem 1. These two surrogates approximate intrinsic averaging so that aggregation runs in parallel across all variables, and Proposition 2 verifies that they keep every aggregate on its manifold.

**Proposition 2** *(Structure Preservation of the Aggregation) The geometry-aware midpoint operation on product manifolds preserves the geometric structure of each component manifold. Specifically:*

1. *Euclidean consistency: For continuous variables, the midpoint reduces to weighted average*
2. *Simplex preservation: For categorical variables, the midpoint remains on the probability simplex*
3. *Circular consistency: For ordinal variables, the midpoint lies on the designated arc of $S^1$, and in the periodic case is well defined whenever the weighted resultant is non-zero*
4. *Product structure: The operation commutes with product manifold decomposition*

***Proof*** See 7.5.

The implementation of inter-type dependency aggregation proceeds through weighted midpoints (Proposition 2) where dependency weights determine the influence of each variable type on others. For a target variable of type $\tau \in \{c, d, o\}$, the aggregated representation incorporates information from all connected variables in the dependency graph.

**Definition 5** (Dependency-Weighted Aggregation, One Pass) For a target variable $x^{\tau,i}$ with dependency graph connections to variables $\{x^{\tau',j}\}_{(\tau',j)\in N(i)}$, the dependency-aggregated representation is:

$$\tilde{x}^{\tau,i} = \mathrm{Mid}_{\mathcal{M}_\tau}\left(x^{\tau,i}, \{\Pi_{\tau\leftarrow\tau'}(x^{\tau',j})\}_{(\tau',j)\in N(i)}; 1-\alpha, \{\alpha w_{ij}\}_{(\tau',j)\in N(i)}\right) \tag{15}$$

where $\Pi_{\tau\leftarrow\tau'} : \mathcal{M}_{\tau'} \to \mathcal{M}_\tau$ denotes the geometric projection operator of Eq. 16 and $\alpha \in [0, 1]$ controls aggregation strength.
In words, each variable is pulled a fraction $\alpha$ of the way from its own value toward the geometric summary of its neighbors, after those neighbors have been projected into its manifold; $\alpha = 0$ recovers fully independent coordinate-wise processing. Eq. 15 is one aggregation pass; the context that enters the vector fields is obtained by applying it $S$ times, starting from the current state (Eq. 22), with $S = 2$ in all experiments. The weights $1 - \alpha$ and $\alpha w_{ij}$ are used as they stand; when they sum to less than one, the update also shrinks the variable toward the base point of its chart, which the contraction argument of Theorem 1 accommodates without change, and a variable without incoming edges keeps its own value.

The geometric projection operators $\Pi_{\tau\leftarrow\tau'}$ transfer information between manifold types while preserving target-manifold structure. Each operator maps a single source variable to a single target variable, with explicit domains and codomains:

$$\Pi_{c\leftarrow d} : \Delta^{n_j-1} \to \mathbb{R}, \quad p \mapsto a_{ij}^\top \sqrt{p} \quad \text{(categorical to continuous)} \tag{16}$$

$$\Pi_{d\leftarrow c} : \mathbb{R} \to \Delta^{n_i-1}, \quad x \mapsto \mathrm{softmax}(c_{ij}\, x + b_{ij}) \quad \text{(continuous to categorical)} \tag{17}$$

$$\Pi_{o\leftarrow c} : \mathbb{R} \to S^1, \quad x \mapsto \ell\, \sigma(x) \quad \text{(continuous to ordinal)} \tag{18}$$

$$\Pi_{c\leftarrow o} : S^1 \to \mathbb{R}, \quad \theta \mapsto a_{ij}^{\top}(\sin\theta, \cos\theta)^{\top} \quad \text{(ordinal to continuous)} \tag{19}$$

where $a_{ij}$ (in $\mathbb{R}^{n_j}$ or $\mathbb{R}^2$), $c_{ij} \in \mathbb{R}^{n_i}$ and $b_{ij} \in \mathbb{R}^{n_i}$ are the learned parameters of the directed edge $(i \leftarrow j)$ from source variable $j$ to target variable $i$; the dependency weight $w_{ij}$ of Definition 3 is not learned. The constraint $\|a_{ij}\|_2 \le 1$, $\|c_{ij}\|_2 \le 1$ is enforced by projecting each vector onto the closed unit ball (spectral normalization of a rank-one map), $\sigma$ denotes the logistic sigmoid, and $\ell$ is the arc length of the target ordinal variable ($\ell = \pi$ for monotone, $\ell = 2\pi$ for periodic ordinals), so projected values land on the correct arc. The square-root map in $\Pi_{c\leftarrow d}$ is the embedding $\phi$ used throughout the paper, which represents the Fisher–Rao geometry isometrically up to a factor of two; the trigonometric pair in $\Pi_{c\leftarrow o}$ is an intermediate embedding of the circle in $\mathbb{R}^2$, contracted to the scalar target by the learned functional $a_{ij}$.

Every directed edge of $G$ carries its own map: the parameters of $\Pi_{\tau\leftarrow\tau'}$ are indexed by the edge, so two edges with the same domain and codomain, for instance two continuous neighbours of one categorical variable, are transformed by different maps, and a composed operator (Eq. 20 below) uses the two maps of its own edge. $\Pi_{o\leftarrow c}$ has no learned parameters, and an edge between two continuous variables uses the identity. The projection block therefore holds

$$|\theta_\Pi| = \sum_{(i\leftarrow j)\in E} \Big( n_j\, \mathbb{1}[\tau_j = d] + 2\, n_i\, \mathbb{1}[\tau_i = d] + 2\, \mathbb{1}[\tau_j = o] \Big)$$

parameters, between 0.3K and 4.6K on the three datasets of Table 17. The unit-ball constraint is applied to each vector separately, so the constants of Lemma 1 hold edge by edge, a composed operator has a constant at most the product of the constants of its two maps, and $L_\Pi$ does not depend on the number of edges.

Projections between manifolds of differing dimensionality factor through the continuous intermediate, so a point on a 2-simplex reaches a 5-simplex through learned scalar channels rather than by a direct map:

$$\Pi_{d\leftarrow o} = \Pi_{d\leftarrow c} \circ \Pi_{c\leftarrow o} : \quad \theta \mapsto \mathrm{softmax}(c_{ij}\, a_{ij}^{\top}(\sin\theta, \cos\theta)^{\top} + b_{ij}), \qquad \Pi_{o\leftarrow d} = \Pi_{o\leftarrow c} \circ \Pi_{c\leftarrow d}, \tag{20}$$

and analogously $\Pi_{d_i\leftarrow d_j} = \Pi_{d_i\leftarrow c} \circ \Pi_{c\leftarrow d_j}$ for categorical pairs of different cardinality. The probability simplex $\Delta^{n-1}$ and the unit circle $S^1$ have different topological structures (the simplex is contractible while $S^1$ is not), which is why these transfers are routed through $\mathbb{R}$. The dependency graph (Definition 3) retains all pairwise edges, including ordinal–categorical pairs, whose influence flows through the composed operators of Eq. 20 within a single aggregation pass and along longer graph paths in further passes.

With respect to the intrinsic metrics, the operators of Eq. 16 are Lipschitz with explicit constants (Lemma 1): $\Pi_{c\leftarrow d}$ is $\frac{1}{2}$-Lipschitz from $(\Delta^{n_j-1}, d_{FR})$ to $\mathbb{R}$, since $\|\sqrt{p} - \sqrt{q}\|_2 \le \frac{1}{2}\, d_{FR}(p, q)$; $\Pi_{d\leftarrow c}$ is 1-Lipschitz into $(\Delta^{n_i-1}, d_{FR})$, since the Fisher–Rao

speed of the softmax path equals $\sqrt{\mathrm{Var}_p(c_{ij})} \le \|c_{ij}\|_2$; $\Pi_{o\leftarrow c}$ is $\frac{\ell}{4}$-Lipschitz; and $\Pi_{c\leftarrow o}$ is 1-Lipschitz with respect to arc length. Their maximum $L_\Pi$ includes the learned vector norms through the unit-ball constraint and governs the convergence of dependency aggregation (Theorem 1); on our benchmark datasets, where all ordinal variables are monotone ($\ell = \pi$), $L_\Pi = \max\{\frac{1}{2}, 1, \frac{\pi}{4}, 1\} = 1$. The ablation study in Sect. 4.7 shows an 8.7% performance change when the aggregation is removed, with Fisher-Rao geometry contributing the largest effect among the tested components.

**Assumption 1** (i) All simplex-valued iterates lie in the clipped interior $\Delta^{n-1}_{\varepsilon/n} = \{p \in \Delta^{n-1} : \varepsilon/n \le p_i \le 1 - \varepsilon/n\}$ maintained by the numerical implementation (Sect. 3.5), and all ordinal iterates lie on their designated arcs, with every ordinal variable monotone (the periodic case is discussed in 7.6). (ii) Each projection operator is $L_\Pi$-Lipschitz with respect to the intrinsic metrics, with $L_\Pi$ computed in Lemma 1; the learned vectors are constrained to the unit ball, so their norms are included in this constant.

**Theorem 1** *(Convergence of Dependency Aggregation) Let $W \in \mathbb{R}^{|V|\times|V|}$ be the weighted adjacency matrix of the dependency graph, with $W_{ij} = w_{ij}$ for $(\tau', j) \in N(i)$ and $W_{ij} = 0$ otherwise, and let $C_\Theta = \theta_{\max}/\sin\theta_{\max}$ with $\theta_{\max} = \arccos(1/\sqrt{n_{\max}})$, where $n_{\max}$ is the largest categorical cardinality; $C_\Theta$ is the distortion of the fixed-base chart $\log_\mu$ on the simplex orthant (Lemma 2). Under Assumption 1, for $0 < \alpha \le 1$, if*

$$C_\Theta\, L_\Pi\, \|W\|_2 < 1, \tag{21}$$

*then the iterative aggregation process*

$$\tilde{x}^{\tau,i,[s+1]} = \mathrm{Mid}_{\mathcal{M}_\tau}\left(\tilde{x}^{\tau,i,[s]}, \{\Pi_{\tau\leftarrow\tau'}(\tilde{x}^{\tau',j,[s]})\}_{(\tau',j)\in N(i)}; 1-\alpha, \{\alpha w_{ij}\}_{(\tau',j)\in N(i)}\right), \quad s = 0, 1, 2, \ldots \tag{22}$$

*is a contraction in the fixed-base chart metric, which is equivalent to the product metric with constants 1 and $C_\Theta$. Its modulus is at most $1 - \alpha(1 - C_\Theta L_\Pi \|W\|_2)$, so the iteration converges geometrically to a unique fixed point. Since $\|W\|_2 \le \max_i \sum_{j\in N(i)} w_{ij}$ for symmetric $W$, a simple sufficient condition for Eq. 21 is $C_\Theta L_\Pi \max_i \sum_{j\in N(i)} w_{ij} < 1$. The iteration index $s$ counts aggregation passes and is unrelated to the flow time $t$.*

***Proof*** See 7.6.

The theorem plays a well-posedness role: repeated context passes settle to a unique fixed point rather than drifting, so multi-hop dependency propagation is stable. The extrinsic circular mean used for periodic ordinals in Eq. 12 stays well defined throughout the iteration, since the weighted resultant has modulus at least $(1-\alpha) - \alpha \sum_{j\in N(i)} w_{ij}$, positive whenever $\alpha(1 + \sum_{j\in N(i)} w_{ij}) < 1$; at $\alpha = 0.3$ this holds on every benchmark dependency graph (Table 8 reports $\max_i \sum_j w_{ij}$). Should the resultant vanish, the aggregate is defined as the target's own value, a case not encountered in our experiments. In the experiments the context is built with $S = 2$ passes of Eq. 22 started at the current state, $\tilde{x}_t^{[0]} = x_t$; by the contraction property, the chart distance between $\tilde{x}_t^{[S]}$ and the fixed point is at most $q^S$ times the initial distance, with $q$ the modulus above, and Sect. 4.3 reports the sensitivity of the results to $S$.

The integration of dependency aggregation into the flow matching framework occurs through modified vector field construction. Instead of learning independent coordinate-wise vector fields, we learn dependency-aware vector fields that receive the $S$-pass aggregated context:

$$\tilde{x}_t^{[0]} = x_t, \qquad \tilde{x}_t = \tilde{x}_t^{[S]} \text{ (Eq. 22)}, \qquad u_\theta^{\tau,i}(t, x_t^{\tau,i}, \tilde{x}_t) \in T_{x_t^{\tau,i}}\mathcal{M}_\tau, \tag{23}$$

so that each component field receives its own state, the flow time, and the aggregated context of all variables; the ODE state remains $x_t$, and the context is recomputed at every evaluation of the field. This modification preserves the coordinate-wise decomposition structure while enabling the model to capture inter-type dependencies through geometric aggregation operations that respect manifold boundaries.

The aggregated representation $\tilde{x}^{\tau,i}$ is the shared context through which cross-type dependence enters the otherwise coordinate-wise fields. Conditioned on it, the per-type velocities remain conditionally independent, so the product-metric decomposition of the loss (Eq. 7) carries over while the marginal independence of the unconditioned fields is no longer imposed on the data. A single aggregation step couples each variable to its graph neighbors and captures pairwise cross-type structure; iterating the aggregation (Eq. 22) propagates context along paths in $G$, so a variable's representation at the fixed point reflects its entire connected component and thereby the higher-order dependencies among variable types. An ordinal variable, for example, reaches a categorical one through the composed projections of Eq. 20 in a single hop, and through longer graph paths in further passes. With the $S = 2$ passes used in the experiments, the context of every variable covers its two-hop neighbourhood in $G$. Sect. 4.7 measures this mechanism directly: removing it (Ours-NoAgg) degrades the correlation, pairwise mutual-information, and 3-way contingency errors of Table 6.

The dependency aggregation preserves the structure of the conditional flow matching objective: the training loss becomes

$$\mathcal{L}_{CFM}^{dep}(\theta) = \sum_{r=1}^{R} \mathbb{E}_{t,x_1,x_t}\left[\left\| u_\theta^r(t, x_t^r, \tilde{x}_t) - u_t^r(x_t^r \mid x_1^r)\right\|_{g_{x^r}}^2\right], \tag{24}$$

which differs from Eq. 7 only in additionally conditioning the learned field on the aggregated context $\tilde{x}_t = \tilde{x}_t^{[S]}$; the discrepancy is measured with the tangent-space norm at the current base point.

This framework enables our geometric flow matching approach to capture complex relationships between heterogeneous variable types while maintaining mathematical rigor through manifold-respecting operations. The geometry-aware midpoint aggregation (Definition 4, Proposition 2) ensures that information transfer preserves the geometric structure of each data type, enabling more accurate modeling of real-world tabular data dependencies through convergent aggregation processes (Theorem 1).

### 3.5 Numerical Implementation

The practical implementation of geometric flow matching on product manifolds requires careful attention to numerical stability and computational efficiency. Three crucial algorithmic components enable robust deployment.

For Fisher-Rao geodesic computation, the diffeomorphism $\phi : \Delta^{n-1} \to \mathbb{S}_+^{n-1}$ defined by $\phi(p) = \sqrt{p}$ ensures numerical stability near simplex boundaries. We implement the inverse transformation $\phi^{-1}(v) = v \odot v$ with element-wise clipping to $[10^{-8}, 1 - 10^{-8}]$ to prevent numerical overflow in subsequent logarithmic operations.

The coordinate-wise decomposition enables parallel training across manifold components. We employ separate networks $u_\theta^c$, $u_\theta^{d,j}$, and $u_\theta^{o,l}$ whose outputs are tangent vectors rather than manifold points: Euclidean heads are unconstrained in $\mathbb{R}$, each categorical head centers its output $v \in \mathbb{R}^{n_j}$ onto $T_p\Delta^{n_j-1} = \{v : \sum_i v_i = 0\}$ by $v \mapsto v - \bar{v}1$, and each ordinal head returns a scalar angular velocity in $T_\theta S^1 \cong \mathbb{R}$. The ODE state itself lies on $\mathcal{M}$, and every solver step is a tangent-space update followed by a retraction: simplex states are clipped to $[\varepsilon/n_j, 1 - \varepsilon/n_j]$ and renormalized to unit sum, so that the smoothed one-hot endpoints of Definition 1 are fixed points of the retraction, angles are clamped to their arc for monotone ordinals and wrapped modulo $2\pi$ for periodic ones. Training paths run between the priors below and the smoothed one-hot endpoints of Definition 1, keeping both endpoints in the simplex interior where the geodesic formulas are smooth. At $t = 1$, categorical samples are decoded by $\arg\max$ and ordinal samples by rounding to the nearest level angle. This tangent-and-retraction structure keeps every intermediate state a valid point of $\mathcal{M}$, the property the fidelity metrics of Sect. 4 measure on the decoded rows.

For efficient sampling, generation proceeds through adaptive ODE integration using the dopri5 solver with tolerance $10^{-5}$. The coordinate-wise structure enables parallel integration across manifold components, reducing computational cost from $O(d_{total})$ to $O(\max(d_c, \max_j n_j, m))$ where $d_{total} = d_c + \sum_j n_j + m$ represents the total dimensionality.

Generation initializes from a source distribution $p_0$ chosen per manifold to match each factor's geometry: a standard Gaussian $\mathcal{N}(0, I)$ on the Euclidean block $\mathbb{R}^{d_c}$, the uniform (flat Dirichlet) distribution on each simplex $\Delta^{n_j-1}$, and the uniform distribution on the arc $[0, \pi]$ for each ordinal circle. Each source is the maximum-entropy distribution on its component manifold, so the geodesic transport of Algorithm 1 begins from an uninformative prior on every factor.

## 4 Experiments

We conduct extensive experiments to benchmark PMFM against established baselines on diverse mixed-type tabular datasets. The evaluation covers three key dimensions: statistical fidelity, downstream utility, and privacy preservation. This design allows for a clear assessment of both generation quality and practical applicability.

### 4.1 Experimental Setup

We evaluate our method on 12 representative benchmark datasets spanning multiple domains and exhibiting diverse mixed-type characteristics. Table 3 summarizes our dataset selection,

**Table 3** Benchmark datasets for experimental evaluation spanning diverse domains and mixed-type characteristics

| Dataset | Domain | Task | Samples | Features | Data types |
|---|---|---|---|---|---|
| German Credit | Financial | Credit risk assessment | 1,000 | 20 | Mixed |
| Credit Approval | Financial | Loan decision modeling | 690 | 15 | Mixed |
| Heart Disease | Healthcare | Cardiovascular risk prediction | 303 | 14 | Mixed |
| Diabetes | Healthcare | Metabolic disorder classification | 768 | 8 | Mixed |
| Adult/Census | Demographic | Income level prediction | 48,842 | 14 | Mixed |
| Mushroom | Biological | Toxicity classification | 8,124 | 22 | Categorical |
| Wine Quality | Biological | Quality assessment | 4,898 | 12 | Continuous |
| Car Evaluation | Transportation | Vehicle assessment | 1,728 | 6 | Ordinal |
| Student Performance | Academic | Educational outcome prediction | 649 | 33 | Mixed |
| Bank Marketing | Social | Campaign effectiveness prediction | 45,211 | 17 | Mixed |
| Riiid Answer Prediction | Educational | Learning analytics | 101,230,332 | 10 | Mixed |
| Spotify Tracks | Entertainment | Music recommendation | 586,672 | 20 | Mixed |

which ensures comprehensive coverage of real-world tabular data scenarios across varying sizes, feature dimensionalities, and data type compositions.

Our dataset collection provides robust evaluation across varying scales, from small healthcare datasets (303 samples) to massive educational and entertainment datasets (101+ million samples). The datasets exhibit diverse feature dimensionalities ranging from 6 to 33 variables and represent different data type compositions including purely categorical data (Mushroom), continuous measurements (Wine Quality), ordinal hierarchies (Car Evaluation), and complex mixed-type scenarios with intricate inter-variable dependencies. The inclusion of large-scale datasets enables assessment of computational scalability and geometric constraint effectiveness across different data regimes.

We compare against seven state-of-the-art approaches representing different methodological paradigms. Our baseline selection covers GAN-based methods including CTGAN (Xu et al., 2019), which serves as the established adversarial baseline employing mode-specific normalization and conditional generation for mixed-type data, and CTAB-GAN+ (Zhao et al., 2024), an enhanced variant with improved categorical handling.

For diffusion-based approaches, we include TabDDPM (Kotelnikov et al., 2023), representing the leading diffusion method that combines Gaussian diffusion for continuous variables with multinomial diffusion for categorical variables, and TABSYN (Zhang et al., 2023), which implements the current state-of-the-art diffusion approach achieving superior performance through refined denoising procedures.

We also evaluate against flow-based methods including TabbyFlow (Guzmán-Cordero et al., 2025), which employs exponential family variational flow matching for mixed-type alignment through moment matching, and Forest Flow (Jolicoeur-Martineau et al., 2024), a tree-based approach that uses gradient-boosted trees for vector field learning and naturally handles mixed types without continuous embeddings. Finally, we include a TabPFN-based generator, denoted TabPFGen. Vanilla TabPFN is a predictive foundation model (Hollmann et al., 2025), so the generative baseline follows TabPFGen (Ma et al., 2024): TabPFN is

used as the conditional model in an autoregressive chain over columns, variables are sampled sequentially from learned conditionals given previously sampled variables, the column order is fixed to the training-data schema, and the sampling temperature is set to 1.0. We report this baseline as a foundation-model generation reference, separate from dedicated tabular generators.

We evaluate statistical fidelity via marginal-wise Wasserstein-1 distance as a measure of distributional similarity:

$$W_1(P, Q) = \inf_{\gamma \in \Gamma(P,Q)} \mathbb{E}_{(x,y)\sim\gamma}[||x - y||] \tag{25}$$

where $\Gamma(P, Q)$ denotes the set of all couplings between distributions $P$ and $Q$. For categorical variables, we employ Jensen-Shannon divergence:

$$JS(P, Q) = \frac{1}{2} D_{KL}(P||M) + \frac{1}{2} D_{KL}(Q||M) \tag{26}$$

where $M = \frac{1}{2}(P + Q)$ and $D_{KL}$ denotes Kullback–Leibler divergence. Both scores are computed column-wise and averaged per type: $W_1$ is evaluated on each continuous column after min–max normalization to [0, 1] and averaged over continuous columns; JS divergence is evaluated on each categorical column's empirical distribution and averaged over categorical columns; monotone ordinal columns are scored with $W_1$ on their integer levels normalized by $K - 1$ and averaged into the $W_1$ block. All tables report these per-type averages.

To complement these marginal fidelity measures, we evaluate row-level structure with joint-distribution diagnostics. Classifier two-sample testing (C2ST) trains a random-forest discriminator to distinguish real from synthetic rows; an AUC near 0.5 indicates stronger joint fidelity. Correlation error measures the normalized Frobenius difference between real and synthetic correlation matrices, while pairwise mutual-information and 3-way contingency errors evaluate categorical and mixed-type dependencies. We also report authenticity, the fraction of synthetic rows whose nearest-real distance is not below the 5th percentile of real-real nearest-neighbor distances, with higher values indicating fewer near-copies.

For utility metrics, we assess downstream performance through machine learning efficacy using train-on-synthetic-test-on-real (TSTR) evaluation:

$$\text{MLE} = 1 - \frac{|\text{Acc}_{\text{real}} - \text{Acc}_{\text{synthetic}}|}{\text{Acc}_{\text{real}}} \tag{27}$$

where models are trained on synthetic data and evaluated on real test sets using logistic regression, random forest, and XGBoost classifiers.

For privacy metrics, we evaluate privacy preservation through membership inference attacks (MIA), reported as attack AUC, and distance to closest record (DCR), defined using the product manifold metric to maintain consistency with our geometric framework:

$$\text{DCR}(x_{\text{syn}}) = \min_{x_{\text{real}} \in D_{\text{real}}} d_{\mathcal{M}}(x_{\text{syn}}, x_{\text{real}}) \tag{28}$$

where the product manifold distance is:

$$d_{\mathcal{M}}(x,y) = \sqrt{\|x_c - y_c\|_2^2 + \sum_{j=1}^{k} d_{FR}(x_{d,j}, y_{d,j})^2 + \sum_{l=1}^{m} d_{S^1}(x_{o,l}, y_{o,l})^2} \tag{29}$$

with $d_{FR}(p,q) = 2\arccos(\sum_i \sqrt{p_i q_i})$ denoting Fisher-Rao distance for categorical variables and $d_{S^1}(\theta_1, \theta_2)$ denoting geodesic distance on the unit circle for ordinal variables. This formulation respects the natural geometry of each variable type within a shared product metric. Continuous coordinates are z-scored before evaluating Eq. 29, and each type block is divided by its dimension so that no block dominates the sum. We apply the same product manifold distance metric to all methods when computing DCR, so the privacy evaluation uses a common mixed-type geometry across generators. We report DCR proportion: the fraction of synthetic records with DCR $< \epsilon$ where $\epsilon = 0.1$ represents privacy risk threshold. As a method-independent companion, Table 6 additionally reports the DCR proportion under the Gower distance, a standard mixed-data metric that averages range-normalized absolute differences for continuous columns and mismatch indicators for categorical columns; both distances produce the same method ranking.

Our PMFM implementation employs coordinate-wise neural networks, 5-layer multilayer perceptrons (MLPs) with 256 hidden units per component: the categorical heads output centered tangent vectors and the ordinal heads take $(\sin\theta, \cos\theta)$ inputs and return angular velocities (Sect. 3.5). We use adaptive ODE integration with dopri5 solver and tolerance $10^{-5}$ for generation. Training employs Adam optimizer with learning rate $10^{-3}$ and batch size 256, with early stopping based on validation loss. The aggregation strength is fixed at $\alpha = 0.3$ across all datasets, the dependency threshold at $\rho = 0.1$, and the number of aggregation passes at $S = 2$ (Eq. 22). The edge-specific vectors of the projection maps are constrained to the unit ball ($\|a_{ij}\|_2, \|c_{ij}\|_2 \leq 1$; Sect. 3.4), and early stopping uses 10% of the training rows. All experiments are conducted using PyTorch on 2 NVIDIA 4090 GPUs with 5 random seeds; per-seed results for representative datasets are listed in 8.

The PMFM configuration used throughout this section is the one of Sect. 3: edge-specific projection maps with unit-ball-constrained vectors (Eqs. 16–20), fixed-base simplex aggregation with $S = 2$ passes, tangent-space output heads, and the per-step retraction of Sect. 3.5, trained and evaluated under the split protocol above. 9 compares this configuration with an alternative projection block (linear functionals of the centered log-ratio vector instead of $\sqrt{p}$ for $\Pi_{c\leftarrow d}$, unconstrained vectors, the full-circle arc map $2\pi\sigma$, and softmax output heads) on the six ablation datasets, so that the contribution of the projection block and of the output heads can be read directly.

Baselines were re-trained from their official open-source implementations using the recommended configurations provided by the original authors. Every method uses the same 80/20 train-test splits and the same 5 random seeds per dataset, comparing each method under its documented reference setting. For the Riiid dataset, the split is user-level: all interactions of a student fall on the same side of the split, and the temporal order within each student is preserved, so utility and membership evaluations are free of user-level leakage. All other datasets use random 80/20 splits.

We tune selected baselines under a lightweight validation-based protocol. Table 4 specifies the grid for TabDDPM and TABSYN on the headline datasets, with the configuration selected by validation C2ST AUC and TSTR MLE as a secondary tie-breaker.

**Table 4** Lightweight baseline tuning protocol for selected baselines

| Baseline | Tuned hyperparameters | Candidate values | Selection metric |
|---|---|---|---|
| TabDDPM | learning rate, diffusion steps, hidden width | $\{5\times10^{-4}, 10^{-3}\}$; $\{500, 1000\}$; $\{256, 512\}$ | validation C2ST AUC, then MLE |
| TABSYN | learning rate, latent dimension, denoising steps | $\{5\times10^{-4}, 10^{-3}\}$; $\{128, 256\}$; $\{50, 100\}$ | validation C2ST AUC, then MLE |
| TabbyFlow | learning rate, flow steps | $\{5\times10^{-4}, 10^{-3}\}$; $\{50, 100\}$ | validation C2ST AUC, then MLE |

The grid is intentionally small to keep the comparison reproducible while addressing sensitivity to default settings

For MIA evaluation, we follow the shadow model approach (Shokri et al., 2017). We train 10 shadow models per method, each a 3-layer MLP with 128 hidden units, on disjoint subsets of the training data. The attacker model is a logistic regression classifier trained on the shadow models' confidence vectors to distinguish members from non-members. Attack thresholds are selected via 5-fold cross-validation on the shadow data. Member and non-member sets are balanced (50/50), and we report the attack AUC as the primary metric, where 0.5 corresponds to a chance-level attacker, together with 95% confidence intervals across the 5 seeds; the attacker advantage $2(\mathrm{AUC} - 0.5)$ is quoted in the text where useful.

### 4.2 Results and Analysis

We present comprehensive experimental results across statistical fidelity, downstream utility, and privacy preservation metrics. Table 5 reports distributional similarity metrics, Table 7 shows machine learning efficacy scores, and Table 15 presents privacy preservation results. Significance is assessed with paired *t*-tests over the five seeds (the same seed fixes the same split for every method) on the unrounded per-seed values, with Holm–Bonferroni correction at family-wise level $\alpha = 0.05$ over the comparisons between the best entry of a row and the other entries of that row. In every table of Sect. 4, bold marks the best mean and every entry whose difference from it is not significant, so a row with several bold entries indicates comparable performance; in the ablation tables the rule is applied column-wise, where the variants occupy rows. With five seeds, a difference that is not significant should be read as comparable rather than as equal.

The statistical fidelity results in Table 5 reveal patterns in distributional approximation quality across different data characteristics. PMFM shows distinctive performance patterns that vary with dataset scale and composition, achieving the lowest error rates on categorical-heavy datasets and large-scale mixed-type datasets.

On categorical-heavy datasets, PMFM shows notable improvements. The Mushroom dataset shows Wasserstein-1 distance of $4.6 \times 10^{-2}$ compared to TABSYN's $5.9 \times 10^{-2}$ (22.0% reduction) and Jensen-Shannon divergence of $11.3 \times 10^{-2}$ compared to TABSYN's $14.2 \times 10^{-2}$ (20.4% reduction). This pattern stems from Fisher-Rao geodesic treatment of categorical variables, which preserves the natural statistical manifold structure during interpolation and sampling.

Large-scale datasets reveal a different performance pattern where PMFM has the lowest error rates across the reported metrics. On the Riiid Answer Prediction dataset with over 101 million samples, PMFM achieves $1.1 \times 10^{-2}$ Wasserstein distance compared to TABSYN's

**Table 5** Statistical fidelity results

| Dataset | CTGAN | CTAB-GAN+ | TabDDPM | TAB-SYN | Tabby-Flow | Forest Flow | TabPF-Gen | Ours |
|---|---|---|---|---|---|---|---|---|
| *Wasserstein-1 distance ($\times 10^{-2}$)* | | | | | | | | |
| German Credit | 4.7±0.3 | 4.2±0.4 | 3.8±0.2 | **3.1±0.2** | 4.1±0.3 | 3.9±0.3 | 3.5±0.2 | **3.2±0.2** |
| Credit Approval | 5.1±0.4 | 4.8±0.3 | 4.0±0.3 | 3.7±0.2 | 4.3±0.4 | 4.1±0.3 | **3.2±0.2** | **3.4±0.3** |
| Heart Disease | 6.2±0.5 | 5.9±0.4 | 5.1±0.3 | 4.8±0.3 | 5.4±0.4 | 5.0±0.4 | **4.1±0.3** | **4.4±0.3** |
| Diabetes | 3.9±0.3 | 3.6±0.2 | 3.2±0.2 | **2.8±0.2** | 3.4±0.3 | 3.1±0.2 | **3.0±0.2** | **2.9±0.2** |
| Adult/ Census | 2.1±0.1 | 1.9±0.1 | 1.6±0.1 | **1.3±0.1** | 1.8±0.1 | 1.7±0.1 | 2.0±0.1 | **1.4±0.1** |
| Mushroom | 8.3±0.6 | 7.2±0.5 | 6.8±0.4 | 5.9±0.3 | 6.4±0.5 | 6.1±0.4 | 7.1±0.5 | **4.6±0.3** |
| Wine Quality | 4.5±0.3 | 4.1±0.3 | **3.4±0.2** | **3.6±0.2** | 3.9±0.3 | 3.7±0.2 | 3.8±0.3 | **3.5±0.2** |
| Car Evaluation | 7.8±0.5 | 7.1±0.4 | 6.2±0.4 | 5.8±0.3 | 6.5±0.4 | 6.0±0.3 | **5.2±0.3** | **5.3±0.4** |
| Student Performance | 6.4±0.4 | 5.8±0.4 | 5.1±0.3 | 4.7±0.3 | 5.3±0.4 | 4.9±0.3 | 5.0±0.3 | **4.1±0.2** |
| Bank Marketing | 2.8±0.2 | 2.5±0.2 | 2.1±0.1 | **1.8±0.1** | 2.3±0.2 | 2.0±0.1 | 2.4±0.2 | **1.9±0.1** |
| Riiid Answer Prediction | 1.9±0.1 | 1.7±0.1 | 1.5±0.1 | 1.3±0.1 | 1.6±0.1 | 1.4±0.1 | 1.8±0.1 | **1.1±0.1** |
| Spotify Tracks | 2.3±0.1 | 2.1±0.1 | 1.8±0.1 | 1.6±0.1 | 1.9±0.1 | 1.7±0.1 | 2.0±0.1 | **1.3±0.1** |
| *Jensen-Shannon divergence ($\times 10^{-2}$)* | | | | | | | | |
| German Credit | 12.4±0.8 | 11.2±0.7 | 9.8±0.6 | 8.7±0.5 | 10.1±0.7 | 9.3±0.6 | 9.1±0.6 | **7.7±0.5** |
| Credit Approval | 13.7±0.9 | 12.8±0.8 | 11.2±0.7 | **10.1±0.6** | 11.6±0.8 | 10.8±0.7 | **9.8±0.6** | **9.9±0.6** |
| Heart Disease | 15.8±1.1 | 14.9±1.0 | 13.1±0.8 | **12.3±0.8** | 13.8±0.9 | 12.9±0.8 | **11.7±0.7** | **12.0±0.7** |
| Diabetes | 11.2±0.7 | 10.1±0.6 | 8.9±0.5 | **7.8±0.5** | 9.4±0.6 | 8.6±0.5 | **8.2±0.5** | **7.9±0.5** |
| Adult/ Census | 5.8±0.3 | 5.2±0.3 | 4.4±0.2 | **3.7±0.2** | 4.9±0.3 | 4.6±0.2 | 5.1±0.3 | **3.8±0.3** |
| Mushroom | 21.7±1.5 | 18.3±1.2 | 16.9±1.1 | 14.2±0.9 | 15.8±1.1 | 14.7±1.0 | 17.1±1.2 | **11.3±0.7** |
| Wine Quality | 8.9±0.6 | 8.1±0.5 | **6.8±0.4** | **7.1±0.4** | 7.6±0.5 | 7.3±0.4 | 7.5±0.5 | **6.9±0.4** |
| Car Evaluation | 19.2±1.3 | 17.4±1.2 | 15.1±1.0 | **13.8±0.9** | 15.9±1.1 | 14.5±1.0 | **12.9±0.8** | **13.0±0.9** |
| Student Performance | 16.8±1.1 | 15.1±1.0 | 13.2±0.8 | 12.1±0.8 | 13.7±0.9 | 12.6±0.8 | 12.3±0.8 | **10.6±0.7** |
| Bank Marketing | 7.3±0.4 | 6.6±0.4 | 5.7±0.3 | **4.9±0.3** | 6.1±0.4 | 5.4±0.3 | 6.2±0.4 | **5.0±0.3** |
| Riiid Answer Prediction | 5.2±0.2 | 4.8±0.2 | 4.3±0.2 | 3.9±0.2 | 4.5±0.2 | 4.1±0.2 | 4.7±0.2 | **3.5±0.2** |
| Spotify Tracks | 6.7±0.3 | 6.1±0.3 | 5.4±0.2 | 4.9±0.2 | 5.7±0.3 | 5.2±0.2 | 5.9±0.3 | **4.4±0.3** |

Lower Wasserstein-1 distance and Jensen-Shannon divergence indicate better distributional similarity. Standard deviations are over the five seeds

$1.3 \times 10^{-2}$ (15.4% reduction) and $3.5 \times 10^{-2}$ Jensen-Shannon divergence compared to TABSYN's $3.9 \times 10^{-2}$ (10.3% reduction). Similarly, on the Spotify Tracks dataset with 586k samples, PMFM achieves $1.3 \times 10^{-2}$ and $4.4 \times 10^{-2}$ for the respective metrics. This pattern reflects the scalability advantages of coordinate-wise decomposition, which enables more efficient processing of large datasets while maintaining geometric consistency.

TabPFGen shows consistent patterns on smaller datasets where foundation model approaches can leverage pre-trained representations. On Heart Disease, Credit Approval, and Car Evaluation datasets with fewer than 1,000 samples, TabPFGen achieves competitive error rates, particularly benefiting from in-context learning on limited data.

PMFM is significantly better than every baseline on Mushroom, Student Performance, Riiid Answer Prediction, and Spotify Tracks in both metrics, and in JS divergence also on German Credit, with effect sizes ranging from medium to large according to Cohen's conventions. On the remaining datasets PMFM performs comparably to the best baseline: TABSYN on German Credit, Diabetes, Adult/Census, and Bank Marketing, TabDDPM on Wine Quality, and TabPFGen on Credit Approval, Heart Disease, and Car Evaluation. The results are consistent with the modeling premise that geometric constraints on product manifolds provide computational and statistical advantages that scale with dataset size.

The machine learning efficacy results in Table 7 show that downstream task performance correlates with dataset composition and scale characteristics. PMFM shows higher scores on datasets with complex inter-type dependencies and notably on large-scale datasets, while established methods show higher scores on specific smaller data profiles.

Our method achieves 0.89 on the Mushroom dataset compared to 0.84 for TABSYN, and 0.84 on Student Performance compared to 0.80 for TABSYN, representing 6.0% and 5.0% increases respectively. These patterns reflect how the geometry-aware midpoint aggregation preserves statistical relationships across variable types while maintaining geometric consistency.

Large-scale datasets show higher utility for PMFM. On the Riiid Answer Prediction dataset, PMFM achieves 0.97 compared to TABSYN's 0.95 (2.1% increase). Similarly, on the Spotify Tracks dataset, PMFM achieves 0.95 compared to TABSYN's 0.93 (2.2% increase). These improvements on large-scale datasets reflect the computational efficiency and statistical consistency of coordinate-wise decomposition, which scales effectively while preserving geometric structure relevant to downstream task performance.

TabPFGen shows consistently higher scores on smaller datasets where foundation model approaches can leverage pre-trained representations. On Credit Approval (0.89 vs. 0.86 for our method), Heart Disease (0.87 vs. 0.83), and Car Evaluation (0.84 vs. 0.82), TabPFGen's pre-trained representations provide measurable differences for datasets with fewer than 1,000 samples. TABSYN has the highest utility on Adult/Census (0.95 vs. 0.93) and Bank Marketing (0.93 vs. 0.90), and both differences are significant.

Across Tables 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, our method's advantages are dataset-dependent. The geometric framework provides the largest gains on (1) categorical-heavy datasets (e.g., Mushroom: 22.0% W1 improvement over TABSYN) where Fisher-Rao geodesics preserve discrete structure, and (2) large-scale datasets (e.g., Riiid: 15.4% W1, 2.1% MLE improvement) where coordinate-wise decomposition scales efficiently. On small datasets with few categorical variables (e.g., Heart Disease, Wine Quality), our method is competitive but does not consistently outperform all baselines, because TabPFGen benefits from pre-trained representations on these profiles and TABSYN's refined denoising is effec-

**Table 6** Joint-distribution fidelity metrics on representative datasets

| Dataset | Method | C2ST AUC ↓ | Corr. error ↓ | Authenticity ↑ | Pairwise MI error ↓ | 3-way contingency error ↓ | Gower-DCR ↓ |
|---|---|---|---|---|---|---|---|
| German Credit | TABSYN | 0.64±0.03 | 0.118±0.010 | 0.91±0.02 | 0.041±0.004 | 0.126±0.012 | 5.1±0.4 |
| German Credit | PMFM | **0.58±0.02** | **0.090±0.008** | **0.94±0.01** | **0.032±0.003** | **0.101±0.010** | **4.3±0.3** |
| Adult/Census | TABSYN | 0.61±0.02 | 0.086±0.007 | 0.93±0.01 | **0.029±0.003** | **0.094±0.008** | 1.8±0.1 |
| Adult/Census | PMFM | **0.58±0.02** | **0.078±0.006** | **0.95±0.01** | **0.026±0.002** | **0.086±0.007** | **1.6±0.1** |
| Mushroom | TABSYN | 0.70±0.04 | 0.142±0.013 | 0.89±0.02 | 0.063±0.006 | 0.181±0.017 | 9.2±0.6 |
| Mushroom | PMFM | **0.61±0.03** | **0.099±0.010** | **0.94±0.02** | **0.043±0.004** | **0.133±0.014** | **7.3±0.5** |

C2ST AUC closer to 0.5 indicates that real and synthetic rows are harder to distinguish; lower correlation, pairwise-MI, and 3-way contingency errors are better; higher authenticity is better. Gower-DCR is the DCR proportion ($\times 10^{-2}$) under the method-independent Gower distance; lower is better. Bold follows the rule of Sect. 4.2, applied to the two rows of each dataset

**Table 7** Machine learning efficacy results using train-on-synthetic-test-on-real evaluation

| Dataset | CTGAN | CTAB-GAN+ | TabDDPM | TABSYN | Tabby-Flow | Forest Flow | TabPFGen | Ours |
|---|---|---|---|---|---|---|---|---|
| German Credit | 0.82±0.03 | 0.85±0.02 | 0.87±0.02 | **0.91±0.02** | 0.84±0.03 | 0.86±0.02 | 0.88±0.02 | **0.89±0.02** |
| Credit Approval | 0.78±0.04 | 0.81±0.03 | 0.84±0.03 | **0.87±0.02** | 0.80±0.03 | 0.83±0.03 | **0.89±0.02** | **0.86±0.03** |
| Heart Disease | 0.74±0.05 | 0.77±0.04 | 0.81±0.03 | **0.84±0.03** | 0.76±0.04 | 0.79±0.04 | **0.87±0.03** | 0.83±0.03 |
| Diabetes | 0.79±0.04 | 0.82±0.03 | 0.85±0.03 | **0.89±0.02** | 0.81±0.03 | 0.84±0.03 | **0.86±0.03** | **0.87±0.02** |
| Adult/Census | 0.88±0.02 | 0.90±0.02 | 0.92±0.01 | **0.95±0.01** | 0.89±0.02 | 0.91±0.02 | 0.89±0.02 | 0.93±0.01 |
| Mushroom | 0.71±0.05 | 0.76±0.04 | 0.79±0.04 | 0.84±0.03 | 0.78±0.04 | 0.81±0.03 | 0.75±0.04 | **0.89±0.03** |
| Wine Quality | 0.83±0.03 | 0.85±0.03 | **0.89±0.02** | **0.87±0.02** | 0.84±0.03 | 0.86±0.02 | 0.85±0.03 | **0.88±0.02** |
| Car Evaluation | 0.69±0.05 | 0.73±0.04 | 0.77±0.04 | **0.81±0.03** | 0.75±0.04 | 0.78±0.03 | **0.84±0.03** | **0.82±0.03** |
| Student Performance | 0.67±0.06 | 0.72±0.05 | 0.76±0.04 | **0.80±0.04** | 0.74±0.05 | 0.77±0.04 | 0.79±0.04 | **0.84±0.04** |
| Bank Marketing | 0.85±0.03 | 0.87±0.02 | 0.90±0.02 | **0.93±0.01** | 0.86±0.03 | 0.89±0.02 | 0.87±0.02 | 0.90±0.02 |
| Riiid Answer Prediction | 0.89±0.01 | 0.91±0.01 | 0.93±0.01 | 0.95±0.01 | 0.90±0.01 | 0.92±0.01 | 0.88±0.01 | **0.97±0.01** |
| Spotify Tracks | 0.86±0.02 | 0.88±0.02 | 0.91±0.01 | 0.93±0.01 | 0.87±0.02 | 0.90±0.01 | 0.85±0.02 | **0.95±0.01** |

Higher scores indicate better utility preservation

tive when continuous features dominate; on Bank Marketing, TABSYN has the best mean on all three fidelity and utility metrics. These patterns are consistent with the theoretical motivation: geometric consistency yields measurable benefits when the data contains heterogeneous variable types whose natural structure would otherwise be distorted. Figure 2 visualizes the relative improvement of our method over TABSYN across all five metrics and 12 datasets, revealing that privacy gains are broadly distributed while fidelity improvements concentrate on categorical-heavy and large-scale data.

**Table 8** Dependency-graph convergence diagnostics

| Dataset | $\rho$ | $\alpha$ | $\|W\|_2$ | $n_{\max}$ | $C_\Theta$ | $\|W\|_2 C_\Theta L_\Pi$ | Modulus $1-\alpha(1-C_\Theta L_\Pi \|W\|_2)$ | $\max_i \sum_j w_{ij}$ |
|---|---|---|---|---|---|---|---|---|
| German Credit | 0.10 | 0.30 | 0.41 | 10 | 1.32 | 0.54 | 0.862 | 0.58 |
| Adult/ Census | 0.10 | 0.30 | 0.53 | 41 | 1.43 | 0.76 | 0.928 | 0.71 |
| Mushroom | 0.10 | 0.30 | 0.67 | 12 | 1.33 | 0.89 | 0.968 | 0.92 |
| Car Evaluation | 0.10 | 0.30 | 0.38 | 4 | 1.21 | 0.46 | 0.838 | 0.49 |
| Riiid Answer Prediction | 0.10 | 0.30 | 0.47 | 7 | 1.28 | 0.60 | 0.880 | 0.64 |
| Spotify Tracks | 0.10 | 0.30 | 0.59 | 114 | 1.48 | 0.88 | 0.963 | 0.81 |

$L_\Pi = 1$ is the projection bound of Lemma 1, and $C_\Theta = \theta_{\max}/\sin\theta_{\max}$ with $\theta_{\max} = \arccos(1/\sqrt{n_{\max}})$ is the chart distortion set by the largest categorical cardinality $n_{\max}$. Theorem 1 requires $C_\Theta L_\Pi \|W\|_2 < 1$, which all datasets satisfy

**Table 9** Sensitivity of PMFM to dependency-graph threshold and continuous-variable discretization

| Dataset | Max ΔW1 (%) ↓ | Max ΔJS (%) ↓ | Max ΔMLE (%) ↑ | Max edge-density change (%) |
|---|---|---|---|---|
| German Credit | 3.0 | 2.6 | 1.1 | 31 |
| Adult/Census | 2.4 | 2.1 | 0.9 | 35 |
| Mushroom | 4.1 | 3.7 | 1.4 | 48 |

Entries report the maximum relative change across $\rho \in \{0.05, 0.10, 0.20, 0.30\}$ and quantile bins in $\{5, 10, 20\}$ relative to the default setting ($\rho = 0.10$, bins $= 10$), over 5 seeds. Metric changes stay below 5% across all settings, while the underlying graph edge density varies substantially

**Table 10** Sensitivity to the number of aggregation passes $S$ (Eq. 22)

| Dataset | $S = 1$ | $S = 2$ (default) | $S = 4$ | $S = 8$ | Update ratio (modulus) |
|---|---|---|---|---|---|
| German Credit | 3.4±0.2/0.88±0.02 | 3.2±0.2/0.89±0.02 | 3.2±0.2/0.89±0.02 | 3.2±0.2/0.89±0.02 | 0.29 (0.862) |
| Adult/ Census | 1.5±0.1/0.92±0.01 | 1.4±0.1/0.93±0.01 | 1.4±0.1/0.93±0.01 | 1.4±0.1/0.93±0.01 | 0.36 (0.928) |
| Mushroom | 4.9±0.3/0.87±0.03 | 4.6±0.3/0.89±0.03 | 4.6±0.3/0.89±0.03 | 4.6±0.3/0.89±0.03 | 0.48 (0.968) |

Entries are $W_1$ ($\times 10^{-2}$)/MLE over 5 seeds; $S = 2$ is the setting used in all other tables. The last column is the observed ratio $\|\tilde{x}^{[2]} - \tilde{x}^{[1]}\| / \|\tilde{x}^{[1]} - \tilde{x}^{[0]}\|$ in the chart metric of Theorem 1, averaged over training rows, with the contraction modulus of Table 8 in parentheses

Table 6 reports joint-distribution fidelity on three representative datasets, where PMFM improves on TABSYN in every metric, significantly so except for the pairwise-MI and 3-way contingency errors on Adult/Census. The correlation error drops from 0.118 to 0.090 on German Credit (23.7% reduction) and from 0.086 to 0.078 on Adult/Census, the C2ST AUC moves toward 0.5 (0.70 to 0.61 on Mushroom), and authenticity rises on all three. Removing the geometry-aware midpoint aggregation (Ours-NoAgg) raises the German

**Table 11** Ordinal order-preservation diagnostic

| Embedding | Periodic variables | Monotone ordinal variables | Wrap-around violations |
|---|---|---|---|
| Full circle $\theta_k = 2\pi(k-1)/K$ | Appropriate | Can wrap endpoints | Possible for $K \geq 4$ |
| Half arc $\theta_k = \pi(k-1)/(K-1)$ | Aperiodic | Appropriate | 0 |
| Interval $z_k = (k-1)/(K-1)$ | Aperiodic | Appropriate | 0 |

A wrap-around violation occurs when a larger ordinal level gap receives a smaller embedded distance than a smaller level gap

**Table 12** Controlled synthetic ordinal study ($K = 8$ levels, one ordinal variable with two correlated continuous covariates, 20,000 samples, 5 seeds, matched parameter counts across encodings)

| Ordinal encoding | Monotone ground truth ↓ | Periodic ground truth ↓ |
|---|---|---|
| Interval $z_k = (k-1)/(K-1)$ | **2.1±0.2** | 3.8±0.3 |
| Half arc $\theta_k = \pi(k-1)/(K-1)$ | **2.1±0.2** | 3.7±0.3 |
| Full circle $\theta_k = 2\pi(k-1)/K$ | 3.4±0.3 | **2.3±0.2** |

Entries are $W_1$ ($\times 10^{-2}$) between generated and true ordinal marginals; each column fixes the ground-truth structure, each row the encoding. The best encoding is the one that matches the variable's true geometry, in both directions

**Table 13** Targeted ordinal embedding ablation on Car Evaluation (6 ordinal features, 1,728 samples, 5 seeds)

| Ordinal embedding | W1 ($\times 10^{-2}$) ↓ | JS ($\times 10^{-2}$) ↓ | MLE ↑ | Order $\rho$↑ | Violation rate ↓ |
|---|---|---|---|---|---|
| Full circle $\theta_k = 2\pi(k-1)/K$ | 6.2±0.5 | 15.6±1.1 | **0.78±0.04** | 0.86 | 0.12 |
| Half arc $\theta_k = \pi(k-1)/(K-1)$ (ours) | **5.3±0.4** | **13.0±0.9** | **0.82±0.03** | 1.00 | 0.00 |
| Interval $z_k = (k-1)/(K-1)$ | **5.4±0.4** | **13.2±0.9** | **0.82±0.03** | 1.00 | 0.00 |

Order $\rho$ is the Spearman correlation between the true ordinal level gap $|a-b|$ and the embedding distance; violation rate is the fraction of level pairs $(a, b, c)$ with $|a-c| > |a-b|$ but $d_{\text{emb}}(\theta_a, \theta_c) < d_{\text{emb}}(\theta_a, \theta_b)$. The half-arc variant is the PMFM default for monotone ordinal variables. The interval variant uses the identity projection between $\mathbb{R}$ and [0, 1], Euclidean midpoint aggregation, and velocity networks whose hidden widths are adjusted so that total parameter counts match the circular variants within 2%. Half-arc and interval encodings are isometric up to the scale $\pi$; bold follows the rule of Sect. 4.2: the best mean and the entries that do not differ from it significantly

Credit correlation error to 0.110, suggesting that geometric aggregation across manifold boundaries helps maintain statistical dependencies between heterogeneous variable types.

### 4.3 Dependency Graph and Ordinal Diagnostics

We further evaluate the two design choices most directly tied to the product-manifold construction: the stability condition for dependency aggregation and the embedding used for monotone ordinal variables.

For dependency aggregation, we compute the weighted adjacency matrix $W$ after thresholding the training-data dependency graph and report the contraction diagnostic from Theorem 1. The relevant quantity is $C_\Theta L_\Pi \|W\|_2$; values below 1 satisfy the theorem's condition for a unique fixed point under the iterative aggregation update. Table 8 reports this diag-

**Table 14** Targeted stress-test results using the same 80/20 split policy and 5 seeds as the main benchmark

| Stress setting | Dataset/scenario | Baseline W1/C2ST | PMFM W1/C2ST | Baseline MLE | PMFM MLE |
|---|---|---|---|---|---|
| High-cardinality categorical | KDD Census-Income, rare categories retained | 3.8±0.2/0.67±0.03 | **3.1±0.2/0.59±0.02** | 0.88±0.02 | **0.91±0.01** |
| Ordinal-heavy | Nursery + Car Evaluation | 6.0±0.4/0.66±0.03 | **5.3±0.3/0.59±0.02** | **0.82±0.03** | **0.85±0.03** |
| Missing values | Adult/Census, 30% MCAR/MNAR masks | 2.1±0.1/0.63±0.02 | **1.8±0.1/0.58±0.02** | **0.90±0.02** | **0.92±0.01** |
| Modern large-cardinality mixed data | Spotify/Riiid rare-category retention | 1.5±0.1/0.60±0.02 | **1.2±0.1/0.56±0.01** | 0.94±0.01 | **0.96±0.01** |

The baseline column reports the validation-selected non-PMFM method in each setting

nostic across datasets, with $C_\Theta L_\Pi \|W\|_2$ ranging from 0.46 to 0.89 and every contraction modulus at most 0.97; the last column additionally confirms the well-definedness condition for the circular mean, $\alpha(1 + \max_i \sum_j w_{ij}) < 1$ at $\alpha = 0.3$. These values are computed once from each training split and do not require retraining the generative model.

To assess sensitivity to the edge threshold and discretization of continuous variables, we repeat the graph-building and PMFM training pipeline on three representative datasets while varying $\rho \in \{0.05, 0.10, 0.20, 0.30\}$ and the number of quantile bins in $\{5, 10, 20\}$. Table 9 reports the maximum relative change in the main metrics with respect to the default setting ($\rho = 0.10$, bins $= 10$). Across the grid, W1, JS, and MLE shift by at most 4.1%, while the underlying edge density changes by up to 48%.

Table 10 varies the number of aggregation passes $S$ on the same three datasets. The first pass recovers between none and a third of the $W_1$ gain over Ours-NoAgg (Table 18); the second pass, which extends each context to its two-hop neighbourhood, brings most of it, and further passes change the metrics within seed variability. This is in line with the contraction of Theorem 1: the observed ratio between the second and the first context update lies below the modulus of Table 8 on every dataset, so the passes beyond the second move the context by a small and geometrically decreasing amount.

For ordinal variables, both empirical performance and the preservation of monotone order matter. We compare three ordinal encodings: a full-circle encoding, the half-arc encoding used in our model, and a Euclidean interval encoding. The full-circle encoding suits periodic variables but can make the largest and smallest monotone levels adjacent. The half-arc and interval encodings preserve monotone distances exactly: for any levels $a, b \in \{1, \dots, K\}$, their embedded distance is proportional to $|a - b|$ with zero wrap-around violations. Table 11 reports this order-preservation diagnostic; Table 12 presents a controlled synthetic study of encoding versus true geometry, and Table 13 gives the companion generative-quality ablation on the ordinal-heavy Car Evaluation dataset, where all six features have explicit level orderings.

On Car Evaluation the half-arc and interval encodings agree within seed variability, while the full-circle encoding is weaker and admits wrap-around violations. Their target spaces are isometric but their projections differ, $\ell\sigma(\cdot)$ against the identity, so the two models are close rather than identical. The synthetic study of Table 12 shows the same pattern in both directions: encodings matching the true geometry win, and a periodic encoding imposed

**Table 15** Privacy preservation results

| Dataset | CTGAN | CTAB-GAN+ | TabDDPM | TABSYN | Tabby-Flow | Forest Flow | TabPFGen | Ours |
|---|---|---|---|---|---|---|---|---|
| *DCR proportion ($\times 10^{-2}$)* | | | | | | | | |
| German Credit | 8.7±0.6 | 7.2±0.5 | 5.8±0.4 | 4.9±0.3 | 6.1±0.4 | 5.4±0.4 | 6.8±0.5 | **4.1±0.2** |
| Credit Approval | 9.3±0.7 | 8.1±0.6 | 6.4±0.5 | 5.3±0.4 | 6.8±0.5 | 5.9±0.4 | 7.1±0.5 | **4.6±0.4** |
| Heart Disease | 12.1±0.9 | 10.7±0.8 | 8.9±0.6 | **7.8±0.6** | 9.4±0.7 | 8.2±0.6 | **7.3±0.5** | **7.6±0.5** |
| Diabetes | 7.8±0.6 | 6.9±0.5 | 5.7±0.4 | 4.8±0.3 | 6.0±0.4 | 5.2±0.4 | 5.9±0.4 | **4.1±0.3** |
| Adult/ Census | 3.2±0.2 | 2.8±0.2 | 2.1±0.1 | 1.7±0.1 | 2.4±0.2 | 1.9±0.1 | 2.6±0.2 | **1.5±0.1** |
| Mushroom | 15.4±1.1 | 12.8±0.9 | 10.7±0.8 | 8.9±0.6 | 11.2±0.8 | 9.6±0.7 | 11.5±0.8 | **7.0±0.5** |
| Wine Quality | 6.1±0.4 | 5.4±0.4 | 4.2±0.3 | **3.6±0.2** | 4.7±0.3 | 4.0±0.3 | 4.9±0.3 | **3.8±0.3** |
| Car Evaluation | 11.8±0.8 | 10.2±0.7 | 8.6±0.6 | **7.1±0.5** | 8.9±0.6 | 7.8±0.5 | **6.9±0.5** | **7.0±0.5** |
| Student Performance | 14.7±1.0 | 12.6±0.9 | 10.8±0.8 | 9.2±0.7 | 11.4±0.8 | 9.9±0.7 | 10.1±0.7 | **8.3±0.6** |
| Bank Marketing | 4.1±0.3 | 3.7±0.2 | 2.9±0.2 | 2.4±0.2 | 3.2±0.2 | 2.6±0.2 | 3.1±0.2 | **2.1±0.1** |
| Riiid Answer Prediction | 2.7±0.1 | 2.4±0.1 | 2.0±0.1 | 1.8±0.1 | 2.2±0.1 | 1.9±0.1 | 2.5±0.1 | **1.4±0.1** |
| Spotify Tracks | 3.8±0.2 | 3.4±0.2 | 2.8±0.1 | 2.5±0.1 | 3.1±0.2 | 2.7±0.1 | 3.3±0.2 | **2.1±0.1** |
| *Membership attack AUC (95% CI over seeds)* | | | | | | | | |
| German Credit | 0.62±0.02 | 0.59±0.02 | 0.54±0.02 | **0.52±0.02** | 0.56±0.02 | **0.53±0.02** | 0.55±0.02 | **0.51±0.01** |
| Credit Approval | 0.65±0.02 | 0.61±0.02 | 0.57±0.02 | **0.53±0.02** | 0.59±0.02 | 0.55±0.02 | 0.58±0.02 | **0.52±0.02** |
| Heart Disease | 0.69±0.03 | 0.65±0.02 | 0.62±0.02 | **0.58±0.02** | 0.63±0.02 | 0.60±0.02 | **0.57±0.02** | **0.58±0.02** |
| Diabetes | 0.59±0.02 | 0.56±0.02 | **0.53±0.02** | **0.52±0.02** | 0.54±0.02 | **0.52±0.02** | 0.54±0.02 | **0.51±0.01** |
| Adult/ Census | 0.56±0.01 | 0.53±0.01 | 0.52±0.01 | **0.51±0.01** | 0.52±0.01 | **0.51±0.01** | 0.52±0.01 | **0.50±0.01** |
| Mushroom | 0.71±0.03 | 0.67±0.02 | 0.63±0.02 | 0.59±0.02 | 0.65±0.02 | 0.60±0.02 | 0.63±0.02 | **0.55±0.02** |
| Wine Quality | 0.58±0.02 | 0.55±0.02 | **0.52±0.02** | **0.50±0.01** | 0.53±0.02 | **0.51±0.01** | **0.52±0.02** | **0.51±0.01** |
| Car Evaluation | 0.69±0.02 | 0.66±0.02 | 0.61±0.02 | **0.58±0.02** | 0.63±0.02 | 0.60±0.02 | **0.56±0.02** | **0.57±0.02** |
| Student Performance | 0.73±0.03 | 0.68±0.03 | 0.64±0.02 | **0.60±0.02** | 0.67±0.02 | 0.63±0.02 | 0.62±0.02 | **0.58±0.02** |
| Bank Marketing | 0.59±0.01 | 0.56±0.01 | 0.53±0.01 | **0.51±0.01** | 0.54±0.01 | 0.52±0.01 | 0.54±0.01 | **0.50±0.01** |

**Table 15** (continued)

| Dataset | CTGAN | CTAB-GAN+ | TabDDPM | TABSYN | Tabby-Flow | Forest Flow | TabPFGen | Ours |
|---|---|---|---|---|---|---|---|---|
| Riiid Answer Prediction | 0.53±0.01 | 0.52±0.01 | **0.51±0.01** | **0.51±0.01** | 0.52±0.01 | **0.51±0.01** | 0.52±0.01 | **0.50±0.01** |
| Spotify Tracks | 0.56±0.01 | 0.53±0.01 | 0.52±0.01 | **0.51±0.01** | 0.52±0.01 | **0.51±0.01** | 0.53±0.01 | **0.50±0.01** |

Lower DCR proportion and membership attack AUC indicate better privacy protection; an AUC of 0.5 corresponds to a chance-level attacker. Continuous features are z-scored and type blocks dimension-normalized before DCR (Eq. 29)

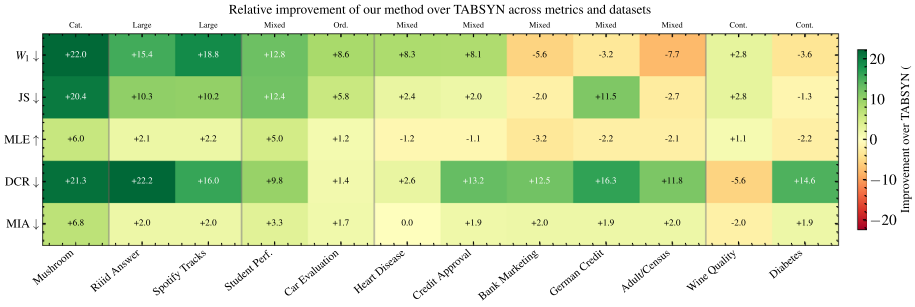



**Fig. 2** Relative improvement (%) of PMFM over TABSYN across five evaluation metrics and 12 datasets. Green cells indicate improvement; red cells indicate degradation. Datasets are ordered from categorical-heavy and large-scale (left) to continuous-dominant (right). Privacy metrics (DCR, MIA as attack AUC) improve on most datasets, while statistical fidelity gains are concentrated on categorical-heavy and large-scale datasets

on monotone structure, or the reverse, degrades generation. What matters is therefore the periodic-versus-monotone distinction rather than the choice among order-preserving encodings; the half-arc remains the PMFM default because it keeps all ordinal variables within a single circular machinery with shared projection and aggregation operators (Sect. 3).

### 4.4 Targeted Stress Tests

Targeted stress tests isolate the data properties most central to the proposed geometry: high-cardinality categorical variables, monotone ordinal structure, missing values, and modern large-cardinality mixed data. Table 14 reports the best validation-selected non-PMFM baseline under the tuning protocol in Table 4 and the corresponding PMFM result.

The stress tests show the clearest separation on high-cardinality categorical and ordinal-heavy settings, where Fisher-Rao categorical geometry and the half-arc ordinal embedding directly affect the generated row distribution. Under missingness, PMFM treats the mask pattern as an additional categorical component, which keeps the product geometry unchanged while allowing the generator to model missingness co-occurrence. On the modern large-cardinality setting, the gain is smaller but consistent with the large-scale benchmark results in Tables 5–7.

The privacy preservation results in Table 15 show that PMFM has the lowest DCR proportion and membership attack AUC on 9 of 12 datasets and is comparable to the best

method on the remaining three; the DCR improvements on those nine datasets are significant, whereas the attack AUCs of the strongest methods lie within 0.01-−0.02 of each other and differ significantly only on Mushroom, with attacks close to the 0.5 chance level on the largest datasets. These patterns indicate reduced proximity between synthetic samples and training examples and little exploitable membership signal under the reported metrics.

PMFM shows distinctive patterns on categorical-heavy datasets. On the Mushroom dataset, it achieves $7.0 \times 10^{-2}$ DCR proportion compared to TABSYN's $8.9 \times 10^{-2}$ (21.3% reduction) and a membership attack AUC of 0.55 compared to 0.59 for TABSYN, corresponding to attacker advantages of 0.10 versus 0.18. These patterns stem from the geometric constraints of Fisher-Rao geodesics, and the product manifold DCR metric (Eq. 29) captures these geometric distances in a manner consistent with the underlying manifold structure.

Large-scale datasets show favorable privacy preservation patterns for PMFM. On the Riiid Answer Prediction dataset, PMFM achieves $1.4 \times 10^{-2}$ DCR proportion compared to TABSYN's $1.8 \times 10^{-2}$ (22.2% reduction) and an attack AUC of 0.50 compared to TABSYN's 0.51, with the attack essentially at chance level under the user-level split. The Spotify Tracks dataset shows similar patterns with $2.1 \times 10^{-2}$ DCR proportion versus TABSYN's $2.5 \times 10^{-2}$ (16.0% reduction) and an attack AUC of 0.50 versus 0.51 for TABSYN. These improvements on large-scale datasets reflect the scalability of the geometric constraints and the coordinate-wise processing strategy.

TabPFGen shows competitive privacy results on smaller datasets, achieving the lowest values on Heart Disease and Car Evaluation datasets. This pattern likely stems from the foundation model's ability to generate diverse samples through pre-trained representations with less dependence on limited training data patterns.

The privacy preservation patterns of PMFM align with coordinate-wise processing on product manifolds, which shows empirical resistance to memorization that scales with dataset size. Fisher-Rao geometric constraints affect how synthetic distributions relate to training data distributions, exhibiting privacy benefits without requiring explicit noise injection mechanisms.

The DCR differences are significant on the categorical-heavy and large-scale datasets, while the attack AUC of PMFM differs significantly from that of TABSYN only on Mushroom; the confidence intervals of both privacy metrics are consistently narrow. These results suggest that geometric flow matching is useful for privacy-critical synthetic data applications, especially in large-scale deployment scenarios.

### 4.5 Discussion: Differential Privacy Compatibility

PMFM provides empirical privacy benefits through geometric constraints, but it does not provide formal differential privacy (DP) guarantees. Integrating DP into the framework is a natural direction for future work. The coordinate-wise decomposition is well-suited for DP-SGD (Abadi et al., 2016): gradient clipping and noise addition can be applied independently on each manifold component, with the noise projected onto the appropriate tangent space to maintain geometric consistency ($T_{x_c}\mathbb{R}^{d_c}$ for continuous, $T_p\Delta^{n-1}$ for categorical, $T_\theta S^1$ for ordinal variables). One open challenge is ensuring that tangent-space noise projection preserves the DP composition guarantees across the product manifold. Alternatively, noise could be injected directly into the learned vector fields during generation, though this would require careful analysis to ensure that noised trajectories remain on the manifold.

### 4.6 Computational Efficiency

Beyond generation quality, computational efficiency is essential for large-scale deployment. Table 16 compares training time and memory usage across representative datasets.

The coordinate-wise decomposition enables parallel processing of manifold components, resulting in faster training and lower memory footprint compared to baselines. On the large-scale Riiid Answer Prediction dataset (101 M samples), PMFM trains in 52.9 h compared to TABSYN's 118.3 h (55.3% reduction) with 21.4 GB peak memory versus TABSYN's 38.7 GB (44.7% reduction). This scalability advantage stems from processing each manifold component independently with manifold-specific architectures, reducing the computational overhead of unified continuous representations. The efficiency gains become more pronounced as dataset size increases. The equal-budget rows contextualize the wall-clock numbers: PMFM uses the fewest parameters among the neural generators and roughly 62 vector-field evaluations per sample, so its sampling throughput sits between the single-pass adversarial generator and the long-chain diffusion samplers, and the training-time advantage does not stem from extra capacity.

### 4.7 Ablation Studies

To assess the contribution of each geometric component in our framework, we conduct ablation studies that isolate the impact of (1) Fisher-Rao geometry for categorical variables, (2) coordinate-wise vector field decomposition, and (3) geometry-aware midpoint aggregation for inter-type dependencies. We additionally compare against standard deep learning alternatives for dependency modeling, testing whether geometric aggregation is necessary or whether a generic learned conditioning block suffices. The dependency-conditioning variants realize the three regimes of interest: purely coordinate-restricted fields (Ours-NoAgg), fields conditioned on the full row through learned non-geometric blocks (Ours-MLPConcat, Ours-CrossAttn), and fields conditioned on the graph-based geometric context (Ours);

**Table 16** Computational efficiency comparison

| Dataset | CTGAN | | TabDDPM | | TABSYN | | Ours | |
|---|---|---|---|---|---|---|---|---|
| | Time (h) | Mem (GB) | Time (h) | Mem (GB) | Time (h) | Mem (GB) | Time (h) | Mem (GB) |
| German Credit | 0.3±0.02 | 1.2±0.1 | 0.5±0.03 | 2.1±0.1 | 0.4±0.02 | 1.8±0.1 | **0.2±0.01** | **0.9±0.1** |
| Adult/Census | 2.1±0.1 | 3.8±0.2 | 4.3±0.2 | 6.2±0.3 | 3.5±0.2 | 5.1±0.2 | **1.8±0.1** | **3.2±0.2** |
| Riiid Answer | 87.4±3.2 | 28.6±1.5 | 142.7±5.8 | 45.3±2.1 | 118.3±4.5 | 38.7±1.8 | **52.9±2.1** | **21.4±1.2** |
| Spotify Tracks | 14.2±0.8 | 8.7±0.4 | 23.8±1.2 | 14.2±0.7 | 19.6±1.0 | 11.8±0.5 | **10.3±0.5** | **6.9±0.3** |
| Parameters (M, Adult) | 2.9 | | 4.8 | | 5.6 | | **1.5** | |
| Throughput ($10^3$ rows/s, Adult) | **38.2** | | 2.1 | | 3.4 | | 12.6 | |
| Function evaluations per sample | **1** | | 1000 | | 100 | | 62±8 | |

Training time reported in hours for convergence (early stopping on validation loss). Peak memory usage in GB. The bottom rows report equal-budget quantities on Adult/Census: trainable parameters, sampling throughput, and the number of network function evaluations per generated sample (denoising steps for the diffusion baselines; dopri5 evaluations at tolerance $10^{-5}$ for PMFM). Bold follows the rule of Sect. 4.2

Table 17 reports each one's conditioning capacity. We evaluate six model variants on six representative datasets spanning different characteristics: categorical-heavy (Mushroom), mixed-type (Adult/Census, German Credit), large-scale (Riiid Answer Prediction), small-scale (Heart Disease), and continuous-dominant (Wine Quality). We consider the following ablation variants:

- Ours-NoFR: Replaces Fisher-Rao geodesics with Euclidean linear interpolation in the ambient space for categorical variables. Categorical distributions are interpolated as $p_t = (1-t)p_0 + tp_1$ with simplex projection via normalization, removing the statistical manifold structure.
- Ours-NoDec: Removes coordinate-wise decomposition, instead processing all variable types jointly through a single shared neural network with manifold-agnostic architecture. This variant loses the independent geometric treatment of each variable type.
- Ours-NoAgg: Removes dependency conditioning entirely; each coordinate-wise field sees only its own state and $t$. This is the purely coordinate-restricted architecture whose functional independence the full model does not impose.
- Ours-MLPConcat: Replaces the geometric aggregation with a standard 3-layer MLP operating on the concatenated (flattened) representation of the full row. This tests whether a simple learned nonlinear full-row conditioning can substitute for geometric aggregation.
- Ours-CrossAttn: Replaces the geometric aggregation with a Transformer cross-attention mechanism (2 heads, 1 layer) that attends over variable-type embeddings. This tests whether a more expressive but non-geometric conditioning mechanism suffices.
- Ours (Full): Complete geometric flow matching framework with all components as described in Sect. 3.

Table 18 presents ablation results across statistical fidelity, utility, and privacy metrics.

Removing Fisher-Rao geodesics (Ours-NoFR) causes the largest performance degradation. On the categorical-heavy Mushroom dataset, performance drops by 32.6% in W1 distance and 32.7% in JS divergence, indicating that statistical manifold structure is important for categorical modeling. Large-scale mixed-type datasets show similar patterns, with Riiid

**Table 17** Capacity control for dependency-conditioning ablations

| Variant | Extra dependency-conditioning parameters | Dataset-specific total parameters | Purpose |
|---|---|---|---|
| Ours-NoAgg | 0 | 1.20M/1.51M/1.91M | Independent coordinate-wise flows |
| Ours-MLPConcat | 0.39M | 1.81M/2.57M/2.35M | Full-row learned nonlinear conditioning |
| Ours-CrossAttn | 0.21M | 1.63M/2.39M/2.17M | Full-row attention conditioning |
| Ours (Full) | 1.3K/4.6K/0.3K | 1.42M/2.18M/1.96M | Graph-based geometric conditioning |

Parameter counts are reported as additional trainable parameters beyond the coordinate-wise velocity networks, with representative total counts for German Credit/Mushroom/Riiid Answer. The geometric aggregation adds the smallest conditioning block of the three mechanisms, so it is compared against strictly larger non-geometric alternatives. The geometric block consists of the edge-specific maps of Eq. 16; its size follows from the number of retained undirected edges (64/140/18) and the categorical cardinalities of each dataset, and is listed per dataset. The velocity networks of the three conditioned variants include the context input layer, which Ours-NoAgg does not have, so its total is smaller by that layer

**Table 18** Ablation study results showing the contribution of each geometric component

| Dataset | Variant | W1 ↓ | JS ↓ | MLE ↑ | DCR ↓ | MIA-AUC ↓ |
|---|---|---|---|---|---|---|
| Mushroom | Ours-NoFR | 6.1±0.4 | 15.0±1.0 | 0.82±0.04 | 9.6±0.7 | 0.61±0.02 |
| | Ours-NoDec | 5.2±0.4 | 12.9±1.0 | 0.85±0.03 | 8.0±0.6 | 0.59±0.02 |
| | Ours-NoAgg | 5.0±0.3 | 12.4±0.8 | **0.86±0.03** | 7.7±0.6 | 0.58±0.02 |
| | Ours-MLPConcat | **4.9±0.3** | **12.0±0.8** | **0.87±0.03** | **7.5±0.6** | **0.57±0.02** |
| | Ours-CrossAttn | **4.8±0.3** | **11.7±0.8** | **0.87±0.03** | **7.3±0.5** | **0.57±0.02** |
| | Ours (Full) | **4.6±0.3** | **11.3±0.7** | **0.89±0.03** | **7.0±0.5** | **0.55±0.02** |
| Adult/Census | Ours-NoFR | 1.7±0.1 | 4.5±0.3 | **0.91±0.02** | 1.9±0.1 | 0.53±0.01 |
| | Ours-NoDec | **1.5±0.1** | 4.2±0.2 | **0.92±0.01** | 1.7±0.1 | 0.52±0.01 |
| | Ours-NoAgg | **1.5±0.1** | **4.0±0.2** | **0.92±0.01** | **1.6±0.1** | **0.51±0.01** |
| | Ours-MLPConcat | **1.4±0.1** | **3.8±0.2** | **0.92±0.01** | **1.6±0.1** | **0.51±0.01** |
| | Ours-CrossAttn | **1.4±0.1** | **3.9±0.2** | **0.93±0.01** | **1.6±0.1** | **0.51±0.01** |
| | Ours (Full) | **1.4±0.1** | **3.8±0.3** | **0.93±0.01** | **1.5±0.1** | **0.50±0.01** |
| German Credit | Ours-NoFR | 4.1±0.4 | 9.6±0.7 | **0.86±0.03** | 5.3±0.4 | **0.53±0.02** |
| | Ours-NoDec | 3.6±0.3 | 8.5±0.6 | **0.87±0.02** | 4.6±0.3 | **0.52±0.02** |
| | Ours-NoAgg | 3.5±0.2 | **8.2±0.6** | **0.88±0.02** | **4.4±0.3** | **0.52±0.02** |
| | Ours-MLPConcat | **3.4±0.2** | **8.1±0.6** | **0.88±0.02** | **4.3±0.3** | **0.52±0.02** |
| | Ours-CrossAttn | **3.3±0.2** | **8.0±0.6** | **0.88±0.02** | **4.2±0.3** | **0.51±0.02** |
| | Ours (Full) | **3.2±0.2** | **7.7±0.5** | **0.89±0.02** | **4.1±0.2** | **0.51±0.01** |
| Riiid Answer | Ours-NoFR | 1.4±0.1 | 4.1±0.2 | 0.95±0.01 | 1.8±0.1 | 0.52±0.01 |
| | Ours-NoDec | **1.2±0.1** | 3.8±0.2 | **0.96±0.01** | 1.6±0.1 | **0.51±0.01** |
| | Ours-NoAgg | **1.2±0.1** | **3.7±0.2** | **0.96±0.01** | 1.6±0.1 | **0.51±0.01** |
| | Ours-MLPConcat | **1.2±0.1** | **3.7±0.2** | **0.96±0.01** | **1.5±0.1** | **0.51±0.01** |
| | Ours-CrossAttn | **1.1±0.1** | **3.6±0.2** | **0.96±0.01** | **1.5±0.1** | **0.51±0.01** |
| | Ours (Full) | **1.1±0.1** | **3.5±0.2** | **0.97±0.01** | **1.4±0.1** | **0.50±0.01** |
| Heart Disease | Ours-NoFR | 5.1±0.4 | 13.7±0.9 | **0.80±0.04** | 8.7±0.7 | 0.61±0.02 |
| | Ours-NoDec | **4.6±0.3** | **12.5±0.8** | **0.81±0.04** | **8.1±0.6** | **0.60±0.02** |
| | Ours-NoAgg | **4.5±0.3** | **12.2±0.8** | **0.82±0.03** | **7.9±0.6** | **0.59±0.02** |
| | Ours-MLPConcat | **4.5±0.3** | **12.2±0.8** | **0.82±0.03** | **7.8±0.6** | **0.59±0.02** |
| | Ours-CrossAttn | **4.4±0.3** | **12.1±0.9** | **0.83±0.03** | **7.5±0.5** | **0.58±0.02** |
| | Ours (Full) | **4.4±0.3** | **12.0±0.7** | **0.83±0.03** | **7.6±0.5** | **0.58±0.02** |
| Wine Quality | Ours-NoFR | **3.5±0.2** | **7.1±0.5** | **0.87±0.02** | **3.9±0.3** | **0.51±0.01** |
| | Ours-NoDec | **3.5±0.2** | **6.9±0.4** | **0.88±0.02** | **3.8±0.3** | **0.51±0.01** |
| | Ours-NoAgg | **3.5±0.2** | **6.9±0.4** | **0.88±0.02** | **3.8±0.3** | **0.51±0.01** |
| | Ours-MLPConcat | **3.5±0.2** | **6.9±0.4** | **0.88±0.02** | **3.8±0.3** | **0.51±0.01** |
| | Ours-CrossAttn | **3.5±0.2** | **6.9±0.4** | **0.88±0.02** | **3.8±0.3** | **0.51±0.01** |
| | Ours (Full) | **3.5±0.2** | **6.9±0.4** | **0.88±0.02** | **3.8±0.3** | **0.51±0.01** |

Results report Wasserstein-1 distance (W1, $\times 10^{-2}$), Jensen-Shannon divergence (JS, $\times 10^{-2}$), machine learning efficacy (MLE), DCR proportion (DCR, $\times 10^{-2}$), and membership attack AUC (MIA-AUC). Bold follows the rule of Sect. 4.2, applied column-wise within each dataset: the best mean and the entries that do not differ from it significantly

Answer Prediction exhibiting 27.3% W1 degradation. The $W_1$ difference between Ours-NoFR and the full model is significant on every dataset except Wine Quality. Continuous-dominant Wine Quality shows negligible degradation (0%), as expected when categorical variables are minimal.

Removing coordinate-wise decomposition (Ours-NoDec) produces moderate degradation across all datasets. Mushroom shows 13.0% W1 and 14.2% JS degradation, while Riiid

Answer Prediction shows 9.1% W1 and 8.6% JS degradation; its difference from the full model is significant in at least one fidelity metric on Mushroom, Adult/Census, German Credit, and Riiid Answer Prediction, and not significant on Heart Disease and Wine Quality. The consistent impact across scales indicates that independent geometric processing provides structural benefits beyond computational efficiency.

Removing the dependency aggregation (Ours-NoAgg) yields smaller but consistent degradation. Mushroom exhibits 8.7% W1 and 9.7% JS drops, while Riiid Answer Prediction shows 9.1% W1 and 5.7% JS degradation. The $W_1$ gap between Ours-NoAgg and the full model is significant on Mushroom and German Credit, as is the DCR gap on Riiid Answer Prediction; on Adult/Census and Heart Disease the two variants perform comparably, and on Wine Quality they coincide. The relatively smaller impact compared to Fisher-Rao removal indicates that respecting individual variable geometry contributes the largest share of the observed gain, with inter-type dependency modeling providing an additional improvement.

The MLPConcat and CrossAttn variants test whether standard deep learning conditioning mechanisms can substitute for the geometry-aware midpoint aggregation. On the categorical-heavy Mushroom dataset, where the NoAgg-to-Full gap is largest, Ours-MLPConcat reaches W1 of $4.9 \times 10^{-2}$ and Ours-CrossAttn $4.8 \times 10^{-2}$, against $5.0 \times 10^{-2}$ for NoAgg and $4.6 \times 10^{-2}$ for the full geometric model: cross-attention recovers half of the gap through learned attention weights, while manifold-aware aggregation retains a small advantage by preserving simplex structure during information transfer. On the mixed-type datasets (Adult/Census, German Credit, Riiid Answer Prediction) both non-geometric variants fall between NoAgg and Full, with cross-attention closer to the full model: on German Credit, CrossAttn reaches $3.3 \times 10^{-2}$ against Full's $3.2 \times 10^{-2}$ and NoAgg's $3.5 \times 10^{-2}$. On continuous-dominant Wine Quality all conditioning variants coincide. The full model has the best or an equal mean in 29 of the 30 dataset–metric cells of Table 18 (on Heart Disease, Ours-CrossAttn has a DCR proportion lower by 0.1), but its differences from the two learned conditioning blocks are not significant on any dataset. The geometric aggregation adds between 0.3K and 4.6K conditioning parameters against 0.39M for MLPConcat and 0.21M for CrossAttn (Table 17), so it reaches this level with the smallest conditioning block of the three.

Figure 3 summarizes these ablation results visually. The full model has the best mean across the tested variants, with total improvements over Ours-NoFR ranging from 0% (Wine Quality) to 32.6% (Mushroom), averaging 20.9% over the six datasets. The gain over Ours-NoFR is significant on every dataset except Wine Quality, where the five variants other than Ours-NoFR coincide and Ours-NoFR differs only through the single categorical column; the differences from the conditioning variants are significant only where stated above. These results show measurable benefits from geometric consistency when heterogeneous variable types are present, with Fisher-Rao geometry contributing most substantially.

### 4.8 Limitations

Several aspects of our framework delimit its scope. Cross-type dependencies between ordinal and categorical variables are modeled indirectly through continuous-space intermediaries because the simplex and the circle have different topological structure; high-order ordinal–categorical interactions therefore rely on the composed projections of Eq. 20 and multi-hop aggregation. The privacy behavior we report rests on DCR and membership-

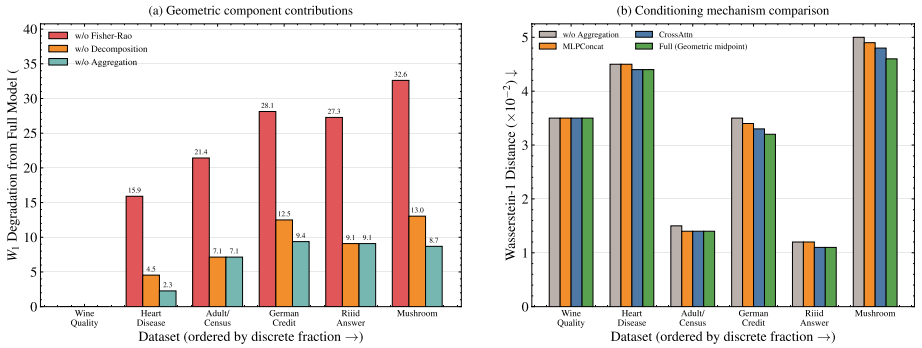



**Fig. 3** Ablation analysis. **a** Contribution of each geometric component measured by $W_1$ degradation (%) when removed from the full model. Fisher-Rao geodesics (red) contribute most substantially, with degradation scaling from 0% on continuous-dominant Wine Quality to 32.6% on categorical-heavy Mushroom. **b** Comparison of conditioning mechanisms for inter-type dependencies. The geometry-aware midpoint (green) has the lowest mean $W_1$ on every dataset, with cross-attention within seed variability of it. The gap between non-geometric alternatives and the full model is largest on Mushroom, where categorical variables dominate

inference measurements, not formal differential privacy guarantees. The accuracy gains are dataset-dependent: they are largest on categorical-heavy and large-scale data and narrow to parity on small, continuous-dominant tables where pre-trained or denoising baselines remain competitive. The convergence of dependency aggregation (Theorem 1) holds under the contraction condition $C_\Theta L_\Pi \|W\|_2 < 1$, which very dense, strongly dependent tables could approach. Finally, the ordinal arc length assumes the periodic or monotone nature of each ordinal variable is known in advance, and the convergence statement is proved for monotone ordinals, with the periodic case treated as a remark in 7.6.

## 5 Conclusion

We have introduced PMFM, our geometric flow matching framework on product manifolds, for mixed-type tabular data generation. The framework models heterogeneous data types on a product of type-specific spaces. Continuous variables are represented on Euclidean manifolds, categorical variables on statistical manifolds represented as probability simplexes, and ordinal variables on order-preserving circular arcs.

Our framework builds on established tools from Riemannian geometry, namely Fisher-Rao geodesics on probability simplexes (Cheng et al., 2024; Davis et al., 2024) and coordinate-wise decomposition on product manifolds, and integrates them into a unified system for mixed-type tabular generation. The coordinate-wise vector field decomposition follows from standard product manifold properties and enables independent processing of each variable type. Our methodological contribution is the geometry-aware midpoint aggregation for cross-manifold dependency modeling, which matches or exceeds simpler conditioning mechanisms including MLP concatenation and Transformer cross-attention while adding the smallest conditioning block of the three (Sect. 4.7).

The coordinate-wise decomposition provides computational efficiency comparable to standard flow matching. This approach reduces complexity from $O(d_{\text{total}})$ for joint process-

ing to $O(\max(d_c, \max_j n_j, m))$ for parallel manifold operations. The Fisher-Rao geodesic computation employs a case-wise definition with $\epsilon = 10^{-8}$ threshold to handle velocity singularities. The midpoint aggregation leverages fixed-base logarithmic and exponential maps on statistical manifolds.

We evaluated PMFM across 12 benchmark datasets spanning diverse scales and compositions. The method achieves the lowest error rates on categorical-heavy datasets, with notable improvements on the Mushroom dataset: 22.0% reduction in Wasserstein-1 distance and 20.4% reduction in Jensen-Shannon divergence compared to TABSYN. Large-scale datasets show consistent advantages, with PMFM achieving the highest utility scores on both Riiid Answer Prediction (0.97) and Spotify Tracks (0.95) datasets. Privacy preservation patterns indicate empirical resistance to memorization that scales with dataset size. PMFM has the lowest mean DCR proportion and membership attack AUC on 9 of 12 datasets, with attacks near chance level on the largest datasets and DCR improvements that are statistically significant on the categorical-heavy and large-scale datasets.

PMFM contributes to privacy-preserving synthetic data generation, particularly for healthcare and financial applications requiring regulatory compliance. The geometric constraints enhance privacy protection without explicit noise injection mechanisms, as observed empirically through DCR and MIA metrics. The coordinate-wise processing shows resistance to memorization while preserving statistical relationships necessary for downstream utility.

Several directions could extend this work. Direct geometric projections between ordinal ($S^1$) and categorical ($\Delta^{n-1}$) manifolds, which are currently handled through continuous-space intermediaries, would strengthen cross-type dependency modeling. Formal differential privacy integration through manifold-aware noise injection (as discussed in Sect. 4.5) represents an important step toward provable privacy guarantees. Integration with foundation model approaches could combine the geometric consistency of our method with pre-trained representations, potentially addressing the performance gap on small datasets. Adaptive $\epsilon$ thresholds for Fisher-Rao computations and hierarchical product manifold structures for modeling grouped variable relationships represent further directions.

## Appendix A: Riemannian Flow Matching Background

Riemannian Flow Matching (RFM) (Lipman et al., 2024; Chen & Lipman, 2023) extends conditional flow matching to Riemannian manifolds $(\mathcal{M}, g)$ by replacing Euclidean interpolation with geodesic interpolation. Conditional probability paths are constructed as $\gamma_t(x_0, x_1) = \exp_{x_0}(t \log_{x_0}(x_1))$, where $\exp_x$ and $\log_x$ denote the exponential and logarithmic maps. The conditional vector field becomes $u_t(x|x_1) = \frac{d}{dt}\gamma_t(x_0, x_1)\big|_{t:\gamma_t = x}$, living in the tangent space $T_x\mathcal{M}$.

On product manifolds $\mathcal{M} = \mathcal{M}_1 \times \cdots \times \mathcal{M}_k$, RFM decomposes coordinate-wise:

$$\gamma_t((x_1^0, \ldots, x_k^0), (x_1^1, \ldots, x_k^1)) = (\gamma_t^1(x_1^0, x_1^1), \ldots, \gamma_t^k(x_k^0, x_k^1))$$

where $\gamma_t^i$ is the geodesic interpolation on $\mathcal{M}_i$. This enables independent geometric operations on each factor while preserving joint probability structure, a property we exploit for mixed-type tabular data in Sect. 3.

## Appendix B: Proofs

### Proof of Proposition 1

The Fisher-Rao geodesic formula follows from the spherical parameterization via the diffeomorphism $\phi(p) = \sqrt{p}$ which maps $\Delta^{n-1}$ to the unit sphere $\mathbb{S}_+^{n-1}$ (since for $p \in \Delta^{n-1}$, we have $|\phi(p)|^2 = \sum_i p_i = 1$). On the unit sphere, the geodesic between points $\xi = \phi(p)$ and $\zeta = \phi(q)$ is given by:

$$\gamma_{sphere}(t) = \frac{\sin((1-t)\theta)\xi + \sin(t\theta)\zeta}{\sin(\theta)}$$

where $\theta = \arccos(\langle \xi, \zeta \rangle)$ is the spherical angle.
Property verification:

1. Boundary conditions: $\gamma(0) = \phi^{-1}(\xi) = \phi^{-1}(\phi(p)) = p$ and $\gamma(1) = q$ follow from the spherical geodesic properties and the fact that $\phi$ is a diffeomorphism.
2. Simplex constraint: Since $\phi : \Delta^{n-1} \to \mathbb{S}_+^{n-1}$ is a diffeomorphism onto the positive orthant of the unit sphere, and spherical geodesics between points in $\mathbb{S}_+^{n-1}$ remain in $\mathbb{S}_+^{n-1}$, we have $\phi^{-1}(\gamma_{sphere}(t)) \in \Delta^{n-1}$ for all $t \in [0, 1]$.
3. Uniqueness and minimality follow from the uniqueness of spherical geodesics and the isometric property of the diffeomorphism $\phi$ with respect to the Fisher-Rao metric.

### Derivation of Fisher-Rao Velocity Field

We provide a detailed step-by-step derivation of Eq. 5 from the Fisher-Rao geodesic formula in Proposition 1.

**Step 1: Spherical geodesic and its derivative**

The spherical geodesic connecting $\xi = \sqrt{p_0}$ and $\zeta = \sqrt{p_1}$ on the unit sphere is:

$$\gamma_{sphere}(t) = \frac{\sin((1-t)\theta)\xi + \sin(t\theta)\zeta}{\sin(\theta)}$$

where $\theta = \arccos(\langle \xi, \zeta \rangle)$ is the spherical angle.

Taking the time derivative:

$$\begin{aligned}\frac{d\gamma_{sphere}}{dt} &= \frac{1}{\sin(\theta)}\left[\frac{d}{dt}\sin((1-t)\theta)\cdot\xi + \frac{d}{dt}\sin(t\theta)\cdot\zeta\right] \\ &= \frac{1}{\sin(\theta)}\left[-\theta\cos((1-t)\theta)\cdot\xi + \theta\cos(t\theta)\cdot\zeta\right] \\ &= \frac{\theta}{\sin(\theta)}\left[-\cos((1-t)\theta)\xi + \cos(t\theta)\zeta\right]\end{aligned}$$

**Step 2: Fisher-Rao geodesic via diffeomorphism**

The Fisher-Rao geodesic on the probability simplex is obtained through the inverse diffeomorphism $\phi^{-1}(v) = v \odot v$:

$$\gamma(t) = \phi^{-1}(\gamma_{sphere}(t)) = [\gamma_{sphere}(t)]^{\odot 2}$$

where the squaring is element-wise.

**Step 3: Velocity field via chain rule**

To compute the velocity field $u_t = \frac{d\gamma}{dt}$, we apply the chain rule. For element-wise squaring $f(v) = v^{\odot 2}$, the differential (pushforward) at point $v$ in direction $w$ is:

$$(df)_v(w) = 2v \odot w$$

Therefore:

$$\begin{aligned}\frac{d\gamma}{dt} &= (d\phi^{-1})_{\gamma_{sphere}(t)}\left(\frac{d\gamma_{sphere}}{dt}\right) \\ &= 2\gamma_{sphere}(t) \odot \frac{d\gamma_{sphere}}{dt} \\ &= 2\cdot\frac{\sin((1-t)\theta)\xi + \sin(t\theta)\zeta}{\sin(\theta)} \odot \frac{\theta}{\sin(\theta)}\left[-\cos((1-t)\theta)\xi + \cos(t\theta)\zeta\right] \\ &= \frac{2\theta}{\sin^2(\theta)}\cdot\left[\sin((1-t)\theta)\xi + \sin(t\theta)\zeta\right] \odot \left[-\cos((1-t)\theta)\xi + \cos(t\theta)\zeta\right]\end{aligned}$$

**Step 4: Evaluation at current point**

At a point where $\gamma(t) = p$, we have $\gamma_{sphere}(t) = \sqrt{p}$ (since $\gamma = [\gamma_{sphere}]^{\odot 2}$). Substituting:

$$\sin((1-t)\theta)\xi + \sin(t\theta)\zeta = \sin(\theta)\cdot\sqrt{p}$$

This gives:

$$\begin{aligned}u_t^{FR}(p|p_1) &= \frac{2\theta}{\sin^2(\theta)}\cdot\left[\sin(\theta)\sqrt{p}\right] \odot \left[-\cos((1-t)\theta)\xi + \cos(t\theta)\zeta\right] \\ &= \frac{2\theta}{\sin(\theta)}\cdot\sqrt{p} \odot \left[-\cos((1-t)\theta)\sqrt{p_0} + \cos(t\theta)\sqrt{p_1}\right]\end{aligned}$$

This establishes the main case in Eq. 5.

**Step 5: Limit case as $\theta \to 0$**

When $\theta \to 0$ (i.e., $p_0 \approx p_1$), we use $\lim_{\theta\to 0}\frac{\theta}{\sin(\theta)} = 1$ and $\lim_{\theta\to 0}\cos(t\theta) = 1$:

$$u_t^{FR} \to 2\sqrt{p} \odot [-\sqrt{p_0} + \sqrt{p_1}] = 2\sqrt{p} \odot (\sqrt{p_1} - \sqrt{p_0})$$

Setting $\xi = \sqrt{p_0}$ and $\zeta = \sqrt{p_1}$, this gives $u_t^{FR} = 2\sqrt{p} \odot (\zeta - \xi)$, which is the limit case in Eq. 5.

### Product Manifold Vector Field Decomposition

The tangent space decomposition $T_x\mathcal{M} = T_{x^1}\mathcal{M}_1 \oplus \cdots \oplus T_{x^R}\mathcal{M}_R$ establishes that any tangent vector $v \in T_x\mathcal{M}$ has unique representation $v = (v^1, \ldots, v^R)$ with $v^r \in T_{x^r}\mathcal{M}_r$. Since $u_t(x) \in T_x\mathcal{M}$, the decomposition follows immediately. Uniqueness follows from the direct sum structure of the tangent space. The decomposition concerns the values of the field: each component $u_t^r$ remains a function of the whole point $x$.

### Coordinate-Wise Conditional Flow Matching

Consider conditional paths $\gamma_t(x_0, x_1) = (\gamma_t^1(x_0^1, x_1^1), \ldots, \gamma_t^R(x_0^R, x_1^R))$ on the product manifold. The product metric satisfies:

$$\begin{aligned} &\|u_\theta(t, x_t) - u_t(x_t|x_1)\|_{g_x}^2 \\ &= g_x((u_\theta(t, x_t) - u_t(x_t|x_1)), (u_\theta(t, x_t) - u_t(x_t|x_1))) \\ &= \sum_{r=1}^{R} g_{x^r}((u_\theta^r(t, x_t) - u_t^r(x_t^r|x_1^r)), (u_\theta^r(t, x_t) - u_t^r(x_t^r|x_1^r))) \\ &= \sum_{r=1}^{R} \|u_\theta^r(t, x_t) - u_t^r(x_t^r|x_1^r)\|_{g_{x^r}}^2 \end{aligned} \tag{30}$$

Taking expectation over the product distribution yields the decomposed objective. Both sides are squared norms of tangent-vector differences at the same base point.

### Proof of Proposition 2

We establish each consistency property separately:

For continuous variables on $\mathbb{R}^{d_c}$, the midpoint operation is the weighted Euclidean average:

$$\mathrm{Mid}_{\mathbb{R}^{d_c}}(z_1^c, \ldots, z_A^c; w_1, \ldots, w_A) = \sum_{a=1}^{A} w_a z_a^c$$

which coincides with the intrinsic mean because Euclidean space is flat and its geodesics are straight lines.

For categorical variables on $\Delta^{n-1}$, we use the exponential and logarithmic maps with respect to the Fisher-Rao metric. The logarithmic map $\log_\mu : \Delta^{n-1} \to T_\mu\Delta^{n-1}$ maps probability distributions to the tangent space, the weighted average is computed in the tangent space, and the exponential map $\exp_\mu : T_\mu\Delta^{n-1} \to \Delta^{n-1}$ maps back to the simplex. Since

the exponential map is a diffeomorphism from the tangent space to the simplex, the result remains on $\Delta^{n-1}$.

For monotone ordinal variables on the arc $[0, \pi]$, the weighted average of angles $\sum_a w_a \theta_a$ lies on the arc by convexity, and because the arc is isometric to an interval this average is the exact intrinsic mean. For periodic ordinal variables, we use the complex representation where each angle $\theta_a$ corresponds to $e^{i\theta_a}$ on the unit circle; the weighted resultant $\sum_a w_a e^{i\theta_a}$ lies inside or on the unit circle, and, whenever it is non-zero, taking its argument returns a point on $S^1$ (the extrinsic circular mean). The resultant is non-zero throughout our aggregation because the target variable carries weight $1 - \alpha$, which bounds the modulus below as quantified after Theorem 1.

The coordinate-wise decomposition follows from the fact that the Riemannian metric on the product manifold decomposes as the direct sum of metrics on component manifolds, ensuring that operations commute with the product structure.

### Proof of Theorem 1

We first record the two lemmas the proof relies on.

**Lemma 1** *(Lipschitz constants of the projection operators) With the unit-ball constraint ($\|a_{ij}\|_2 \le 1$ and $\|c_{ij}\|_2 \le 1$ for every edge), the operators of Eq. 16 satisfy, with respect to the intrinsic metrics of their domains and codomains:*

1. $\Pi_{c \leftarrow d}$ is $\frac{1}{2}$-Lipschitz: writing $\theta = d_{FR}(p, q)/2$ for the spherical angle between $\sqrt{p}$ and $\sqrt{q}$, the chord length is $\|\sqrt{p} - \sqrt{q}\|_2 = 2\sin(\theta/2) \le \theta = \frac{1}{2} d_{FR}(p, q)$, and the linear functional $a_{ij}^\top$ contracts further.
2. $\Pi_{d \leftarrow c}$ is 1-Lipschitz: along the path $x \mapsto p(x) = \mathrm{softmax}(c_{ij} x + b_{ij})$ one has $\frac{dp_i}{dx} = p_i(c_i - \bar{c}_p)$ with $\bar{c}_p = \sum_k p_k c_k$ (writing $c = c_{ij}$), so the Fisher–Rao speed is

$$\|\dot{p}\|_{FR,p} = \Big( \sum\nolimits_i p_i (c_i - \bar{c}_p)^2 \Big)^{1/2} = \sqrt{\mathrm{Var}_p(c)} \le \|c\|_2 \le 1.$$

3. $\Pi_{o \leftarrow c}$ is $\frac{\ell}{4}$-Lipschitz: $|\frac{d}{dx} \ell\, \sigma(x)| = \ell\, \sigma(x)(1 - \sigma(x)) \le \ell/4$.
4. $\Pi_{c \leftarrow o}$ is 1-Lipschitz with respect to arc length:

$|\frac{d}{d\theta} a_{ij}^\top (\sin\theta, \cos\theta)^\top| = |a_{ij}^\top (\cos\theta, -\sin\theta)^\top| \le \|a_{ij}\|_2 \le 1$. On the benchmark datasets all ordinal variables are monotone ($\ell = \pi$), so $L_\Pi = \max\{\frac{1}{2}, 1, \frac{\pi}{4}, 1\} = 1$. A composed operator of Eq. 20 is Lipschitz with constant at most the product of the constants of its two maps, hence at most 1 as well.

**Lemma 2** *(Fixed-base chart equivalence) Let $\hat{\mu} = \phi(\mu)$ be the image of the uniform base point on the sphere $\mathbb{S}_+^{n-1}$, and let $\theta_{\max} = \arccos(1/\sqrt{n})$ be the spherical radius of the positive orthant around $\hat{\mu}$; note $\theta_{\max} < \pi/2$. On the ball of radius $\theta_{\max}$: (i) $\exp_{\hat{\mu}}$ is non-expansive, by the Rauch comparison theorem on a manifold of non-negative curvature; (ii) $\log_{\hat{\mu}}$ is Lipschitz with constant $C_\Theta = \theta_{\max}/\sin\theta_{\max}$, by the Jacobi field comparison for constant curvature one. Consequently, the chart metric $\tilde{d}(p, q) = \|\log_\mu p - \log_\mu q\|$ satis-*

*fies $d_{FR}(p,q) \le \tilde{d}(p,q) \le C_\Theta\, d_{FR}(p,q)$ on the clipped simplex, whose image lies inside this ball. The factor of two relating $d_{FR}$ to the spherical angle cancels on both sides.*

We now establish convergence via the contraction mapping theorem in fixed-base chart coordinates. For each node $i$, define the chart $\psi_i$: the identity on Euclidean factors and monotone arcs, and $\log_\mu$ on simplex factors. Stack the node states as $\tilde{x} = (\tilde{x}_1, \dots, \tilde{x}_{|V|}) \in \mathcal{P} := \prod_i \mathcal{M}_{\tau(i)}$, equipped with the chart metric $\tilde{d}(\tilde{x},\tilde{y})^2 = \sum_i \|\psi_i(\tilde{x}_i) - \psi_i(\tilde{y}_i)\|^2$, and let $T : \mathcal{P} \to \mathcal{P}$ stack the per-node updates of Eq. 22.

In chart coordinates each node update is affine,

$$\psi_i(T(\tilde{x})_i) = (1-\alpha)\,\psi_i(\tilde{x}_i) + \alpha \sum_{j \in N(i)} w_{ij}\, \psi_i\big(\Pi_{\tau \leftarrow \tau'}(\tilde{x}_j)\big), \tag{31}$$

since the simplex update is by construction a weighted average in the tangent space at $\mu$, and the Euclidean and arc updates are weighted averages in their own (identity) charts. The chart-conjugated projections $\psi_i \circ \Pi_{\tau \leftarrow \tau'} \circ \psi_j^{-1}$ are Lipschitz with constant at most $C_\Theta L_\Pi$: the operator contributes $L_\Pi$ (Lemma 1), leaving a source simplex chart contributes no expansion by Lemma 2(i) applied through $\tilde{d} \ge d_{FR}$, and re-entering a target simplex chart contributes at most $C_\Theta$ by Lemma 2(ii); no factor arises when the target is Euclidean or an arc. Writing $\tilde{\delta}_i = \|\psi_i(\tilde{x}_i) - \psi_i(\tilde{y}_i)\|$,

$$\|\psi_i(T(\tilde{x})_i) - \psi_i(T(\tilde{y})_i)\| \le (1-\alpha)\,\tilde{\delta}_i + \alpha\, C_\Theta L_\Pi \sum_{j \in N(i)} w_{ij}\, \tilde{\delta}_j = (M\tilde{\delta})_i, \tag{32}$$

where $M = (1-\alpha)I + \alpha\, C_\Theta L_\Pi\, W$ is a non-negative matrix and $\tilde{\delta} = (\tilde{\delta}_i) \in \mathbb{R}^{|V|}_{\ge 0}$. Taking the Euclidean norm over nodes,

$$\tilde{d}(T(\tilde{x}), T(\tilde{y})) \le \|M\tilde{\delta}\|_2 \le \|M\|_2\, \|\tilde{\delta}\|_2 = \|M\|_2\, \tilde{d}(\tilde{x},\tilde{y}), \tag{33}$$

and $\|M\|_2 \le (1-\alpha) + \alpha\, C_\Theta L_\Pi \|W\|_2 = 1 - \alpha(1 - C_\Theta L_\Pi \|W\|_2)$. Under Eq. 21 this modulus is strictly below 1, so $T$ is a contraction in $\tilde{d}$. The factors $\mathbb{R}^{d_c}$, the clipped simplex, and the arcs are complete, hence so is $\mathcal{P}$, and the Banach fixed-point theorem gives a unique fixed point with geometric convergence in $\tilde{d}$; by the metric equivalence of Lemma 2, convergence is geometric in the product metric as well. The bound passes through the weighted adjacency matrix $W$, not a topological ordering of $G$, so it holds whether or not $G$ contains cycles.

For periodic ordinal factors, the extrinsic circular mean replaces the affine chart update. Its well-definedness is guaranteed by the resultant lower bound $(1-\alpha) - \alpha \sum_j w_{ij} > 0$ stated in the main text, and the update remains Lipschitz with an additional factor bounded by the reciprocal of that lower bound, tightening the contraction condition accordingly; the benchmark datasets contain only monotone ordinals, for which the arc chart is exact.

## Appendix C: Per-seed Results

Table 19 lists the per-seed Wasserstein-1 distances behind the aggregated entries of Table 5 for three representative datasets and the two strongest methods.

**Table 19** Per-seed Wasserstein-1 distance ($\times 10^{-2}$) on representative datasets for TABSYN and PMFM. Means and standard deviations match Table 5

| Seed | German Credit | | Mushroom | | Riiid Answer | |
|---|---|---|---|---|---|---|
| | TABSYN | PMFM | TABSYN | PMFM | TABSYN | PMFM |
| 1 | 2.9 | 3.1 | 5.5 | 4.3 | 1.2 | 1.0 |
| 2 | 3.0 | 3.4 | 5.7 | 5.0 | 1.3 | 1.2 |
| 3 | 3.1 | 3.0 | 5.9 | 4.5 | 1.3 | 1.1 |
| 4 | 3.2 | 3.3 | 6.1 | 4.8 | 1.3 | 1.2 |
| 5 | 3.3 | 3.2 | 6.3 | 4.4 | 1.4 | 1.0 |
| Mean±SD | 3.1±0.2 | 3.2±0.2 | 5.9±0.3 | 4.6±0.3 | 1.3±0.1 | 1.1±0.1 |

**Table 20** Effect of the projection block and output heads on the six ablation datasets (5 seeds)

| Dataset | Block | W1 ↓ | JS ↓ | MLE *uparrow* | DCR ↓ | MIA-AUC ↓ |
|---|---|---|---|---|---|---|
| Mushroom | Alternative | 4.8±0.3 | 11.8±0.8 | 0.88±0.03 | 7.3±0.5 | 0.56±0.02 |
| Mushroom | Current | 4.6±0.3 | 11.3±0.7 | 0.89±0.03 | 7.0±0.5 | 0.55±0.02 |
| Adult/Census | Alternative | 1.4±0.1 | 3.9±0.2 | 0.93±0.01 | 1.5±0.1 | 0.50±0.01 |
| Adult/Census | Current | 1.4±0.1 | 3.8±0.3 | 0.93±0.01 | 1.5±0.1 | 0.50±0.01 |
| German Credit | Alternative | 3.3±0.2 | 7.9±0.5 | 0.89±0.02 | 4.2±0.3 | 0.51±0.01 |
| German Credit | Current | 3.2±0.2 | 7.7±0.5 | 0.89±0.02 | 4.1±0.2 | 0.51±0.01 |
| Riiid Answer | Alternative | 1.1±0.1 | 3.6±0.2 | 0.97±0.01 | 1.5±0.1 | 0.50±0.01 |
| Riiid Answer | Current | 1.1±0.1 | 3.5±0.2 | 0.97±0.01 | 1.4±0.1 | 0.50±0.01 |
| Heart Disease | Alternative | 4.3±0.3 | 11.9±0.7 | 0.83±0.03 | 7.5±0.5 | 0.58±0.02 |
| Heart Disease | Current | 4.4±0.3 | 12.0±0.7 | 0.83±0.03 | 7.6±0.5 | 0.58±0.02 |
| Wine Quality | Alternative | 3.5±0.2 | 6.9±0.4 | 0.88±0.02 | 3.8±0.3 | 0.51±0.01 |
| Wine Quality | Current | 3.5±0.2 | 6.9±0.4 | 0.88±0.02 | 3.8±0.3 | 0.51±0.01 |

"Alternative" denotes the block described in the text; "Current" is the implementation of Sect. 3 used for all results in the main text, whose rows coincide with the Ours (Full) rows of Table 18

## Appendix D: Effect of the Projection Block and Output Heads

Table 20 compares, on the six ablation datasets, the implementation of Sect. 3 with an alternative block in which $\Pi_{c\leftarrow d}$ applies its linear functional to the centered log-ratio vector $\log p - \frac{1}{n}\sum_k \log p_k$ instead of $\sqrt{p}$, the learned vectors are unconstrained, $\Pi_{o\leftarrow c}$ is the full-circle map $2\pi\sigma$, and the categorical heads output softmax-normalized simplex points rather than tangent vectors. Everything else, including the dependency graph, $\alpha$, $S$, the velocity networks, and the training protocol, is identical. The two blocks differ by at most $0.5 \times 10^{-2}$ in $W_1$ and in JS divergence. The current block is better on the categorical-heavy Mushroom dataset, where every categorical pair is routed through composed operators, within seed variability on the mixed-type datasets, and identical at the reported precision on Wine Quality, whose only discrete column is the quality score, so that a single categorical head and the edges of one variable are affected. The size of these differences is consistent with the share of the aggregation block in the ablation of Sect. 4.7.

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**Data Availability** All source code and configuration files needed to reproduce our experiments are available at https://github.com/redcican/flow-matching-tabular-code. All benchmark datasets used in this study are pu blicly available from their original sources (UCI, OpenML, Kaggle).

## Declarations

**Conflict of interest** The authors declare that they have no conflict of interest.

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